

BFS Traversal

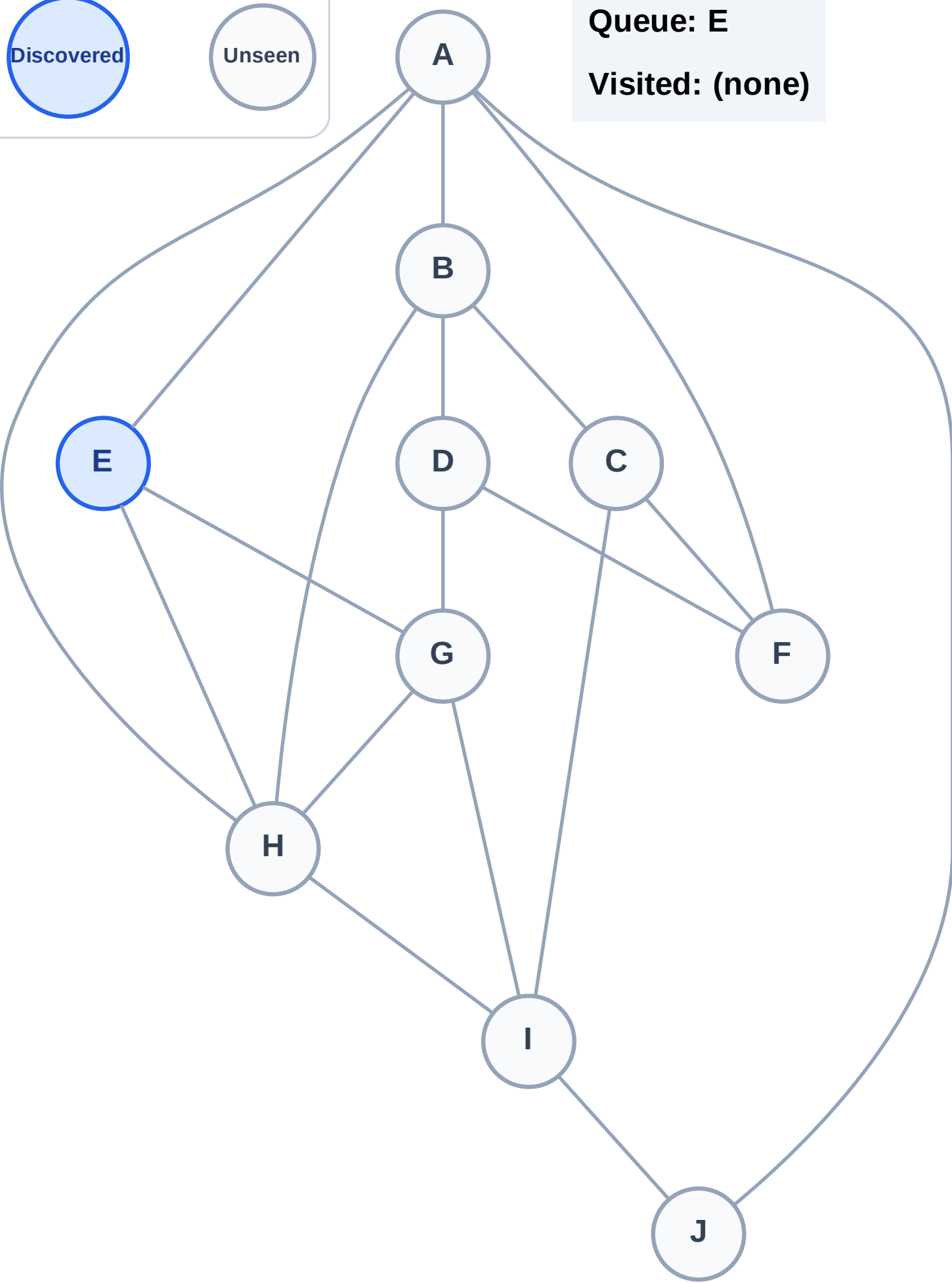
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

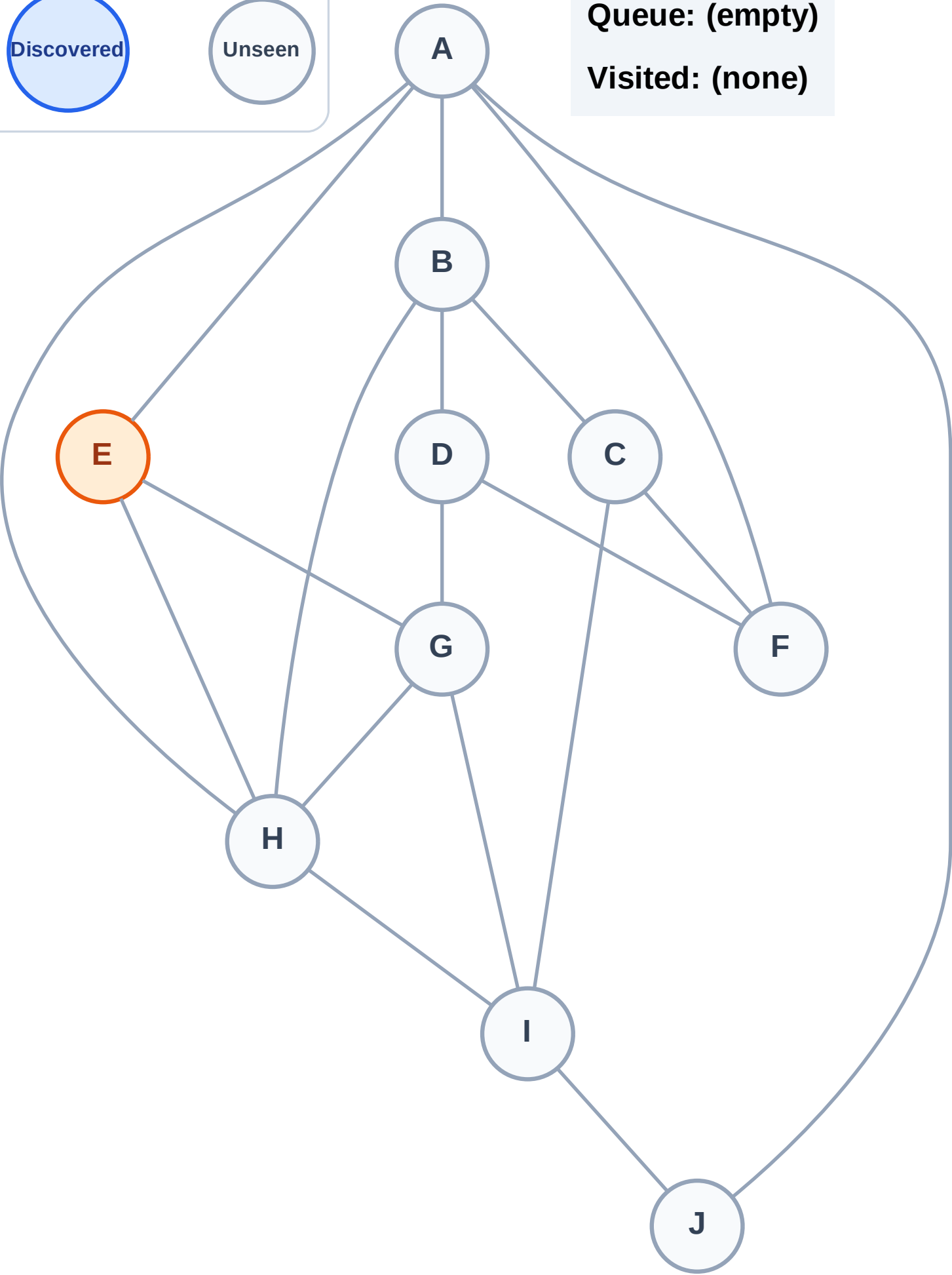
Complete

Processing

Discovered

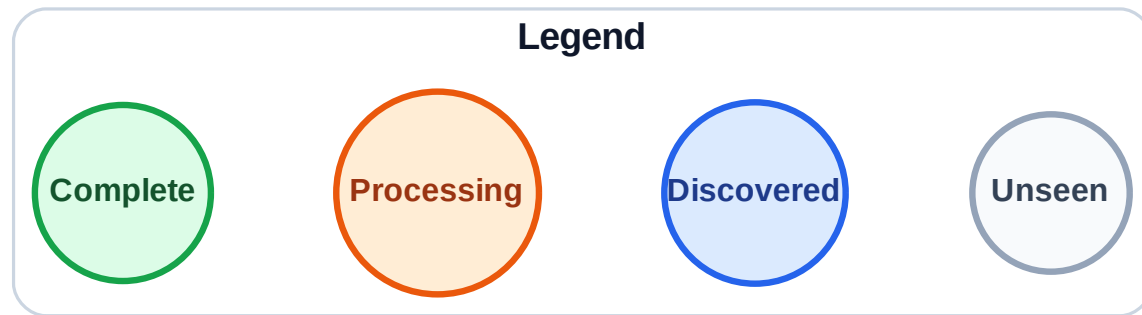
Unseen

Queue: (empty)
Visited: (none)

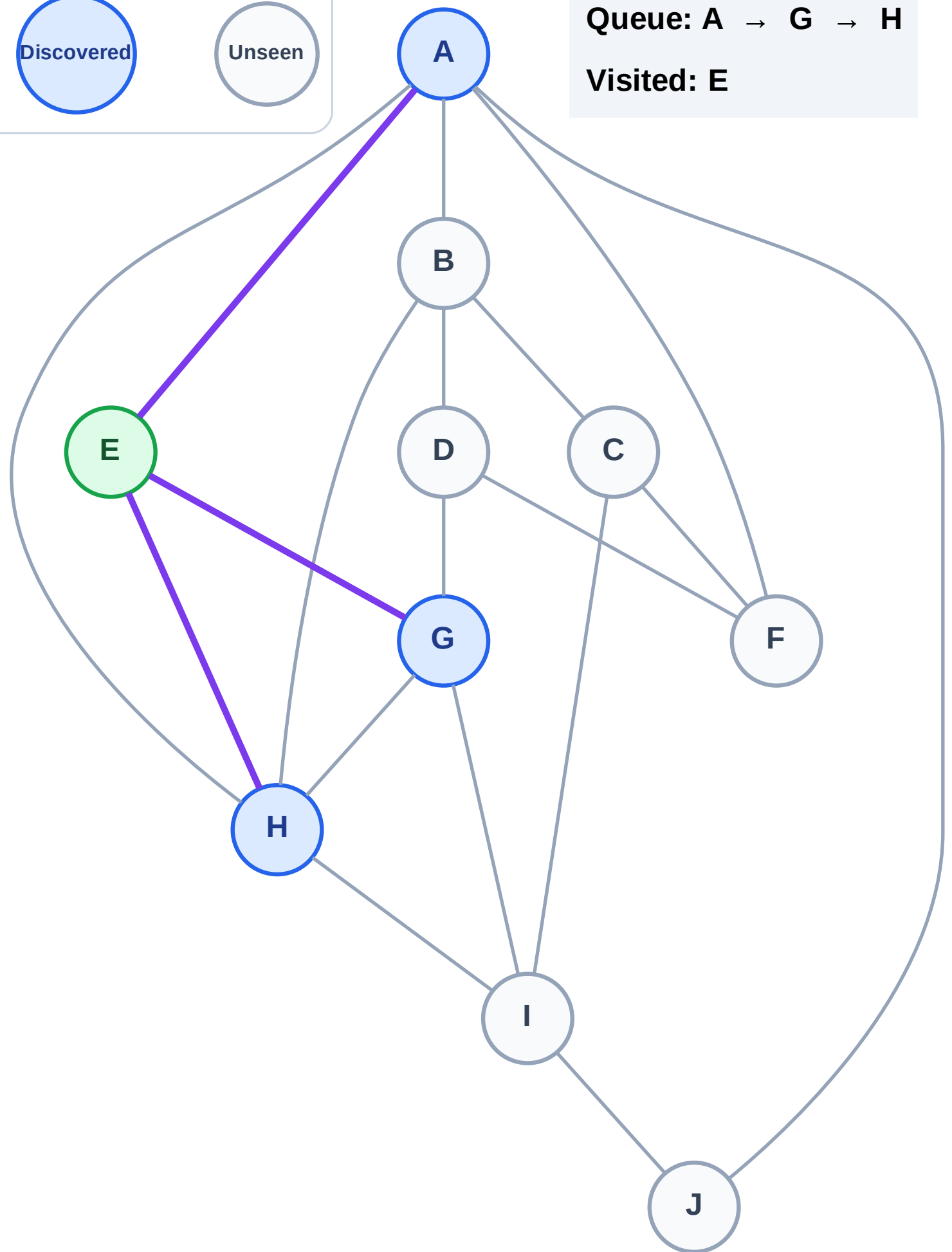


BFS Traversal

Step 3: Discover A, G, H from E



Queue: A → G → H
Visited: E



BFS Traversal

Step 4: Process node A

Legend

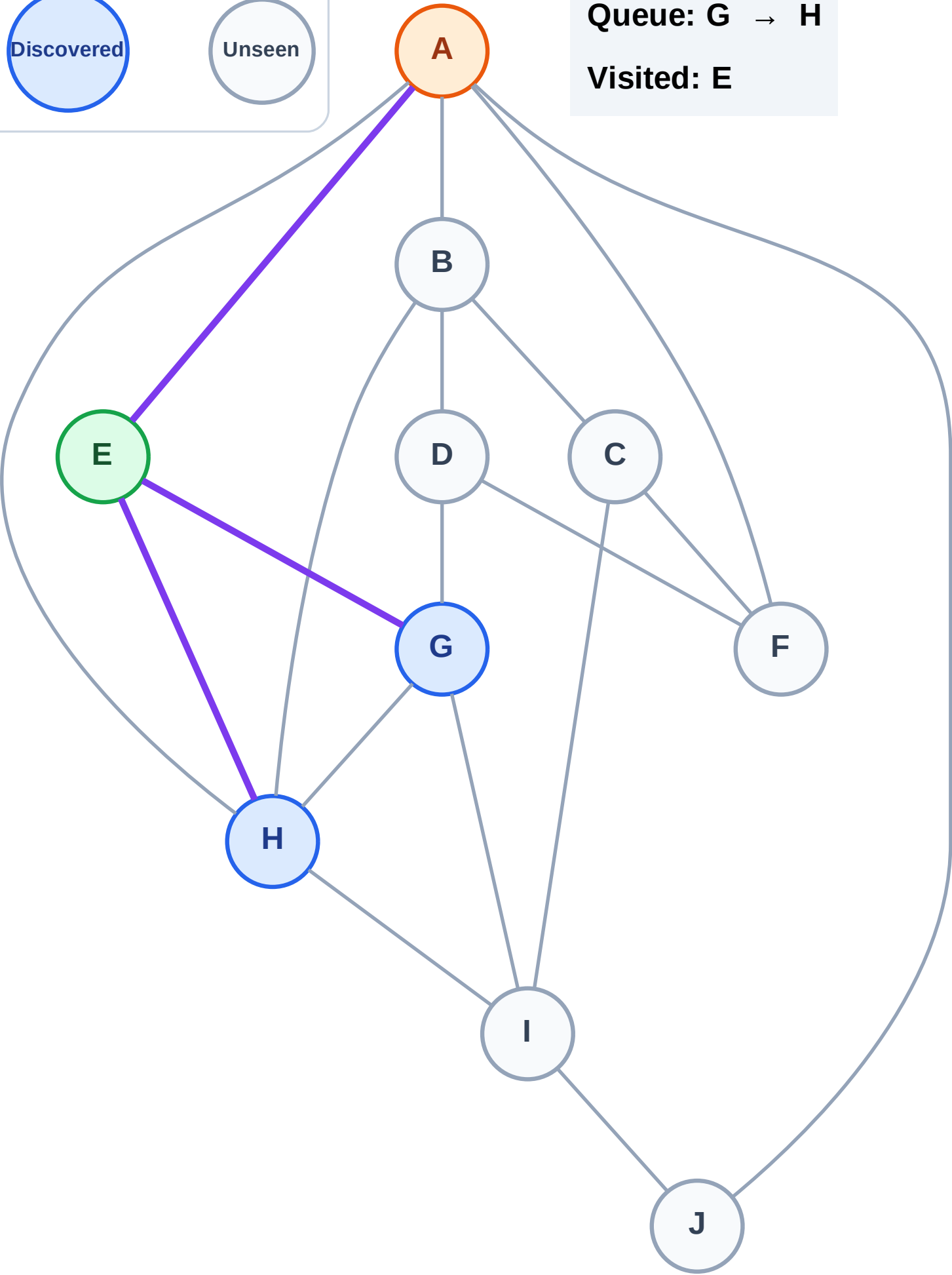
Complete

Processing

Discovered

Unseen

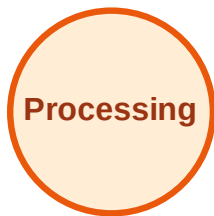
Queue: G → H
Visited: E



BFS Traversal

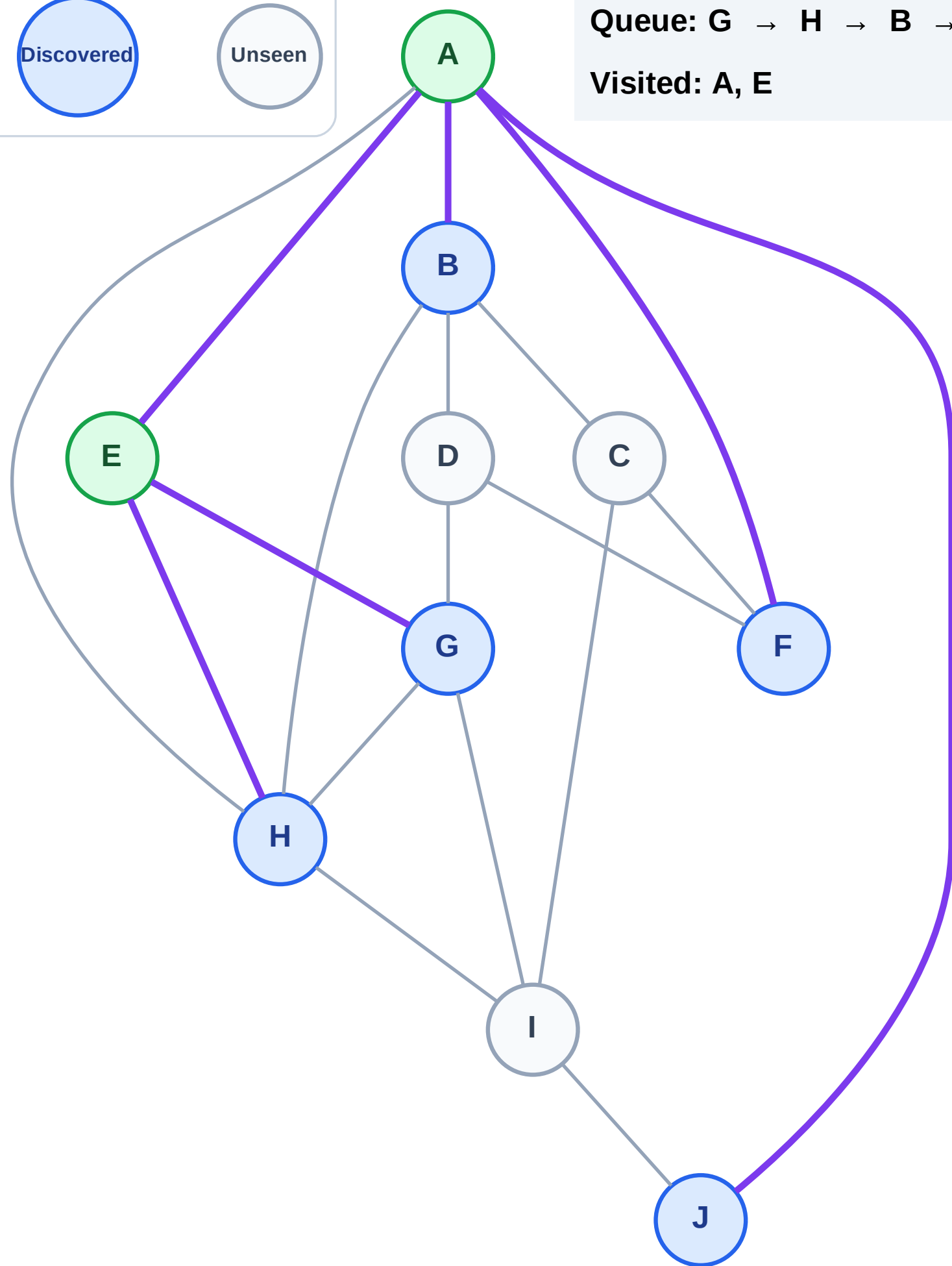
Step 5: Discover B, F, J from A

Legend



Queue: G → H → B → F → J

Visited: A, E



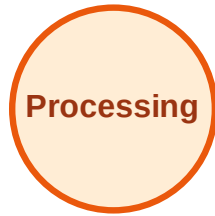
BFS Traversal

Step 6: Process node G

Legend



Complete



Processing

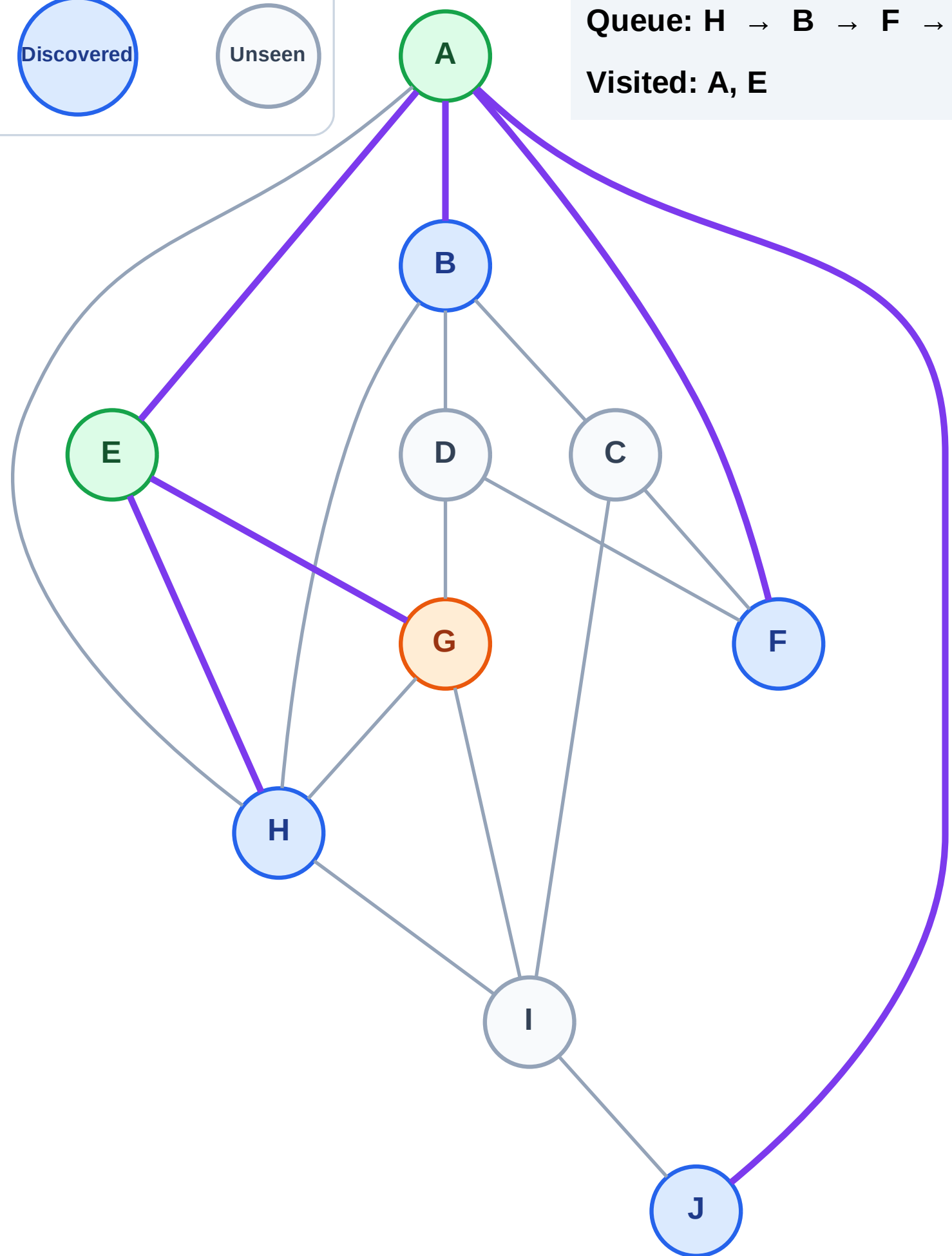


Discovered



Unseen

Queue: H → B → F → J
Visited: A, E



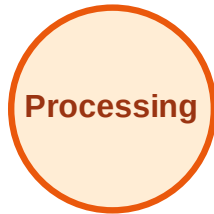
BFS Traversal

Step 7: Discover D, I from G

Legend



Complete



Processing

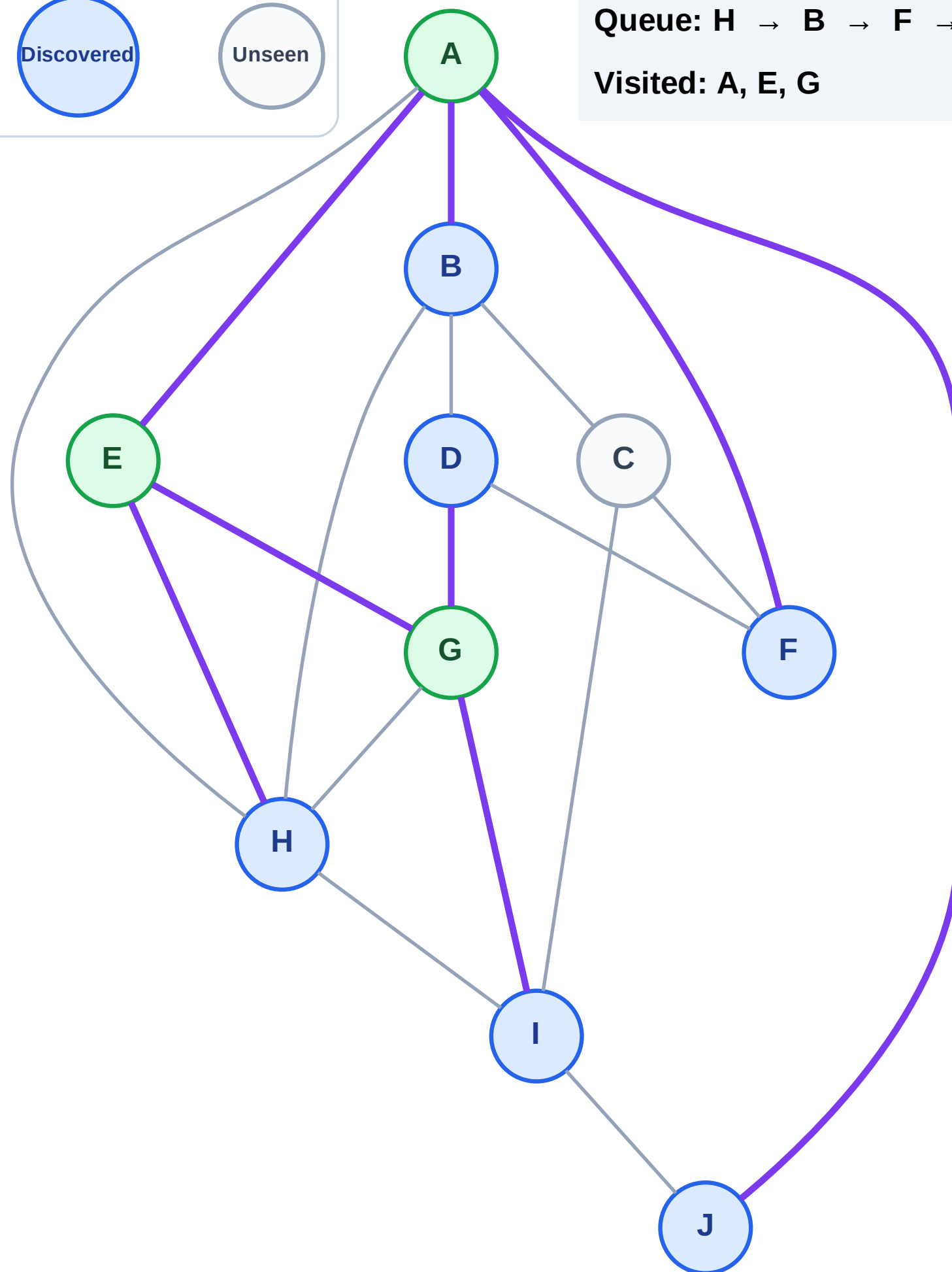


Discovered



Unseen

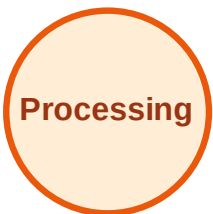
Queue: H → B → F → J → D → I
Visited: A, E, G



BFS Traversal

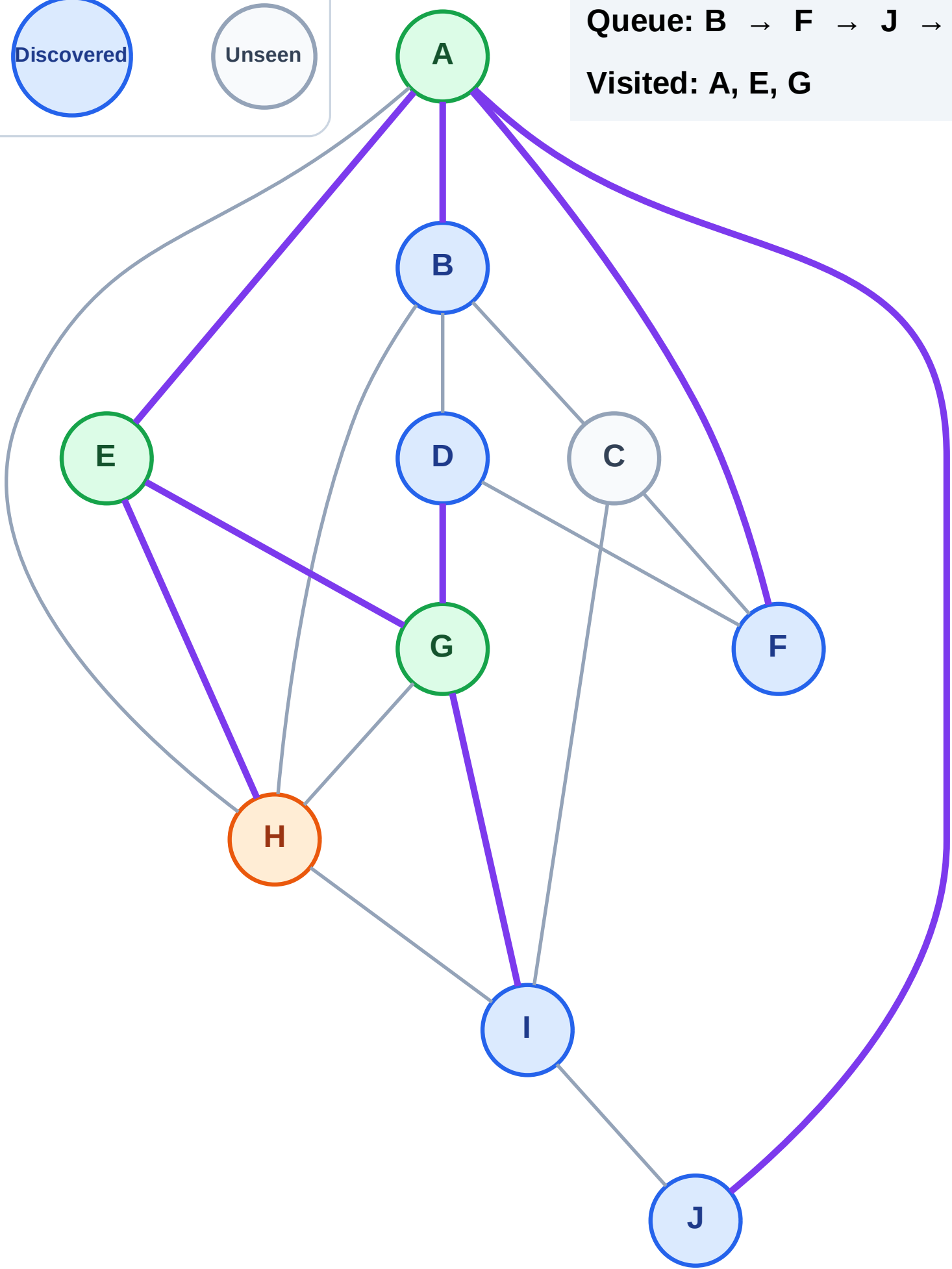
Step 8: Process node H

Legend



Queue: B → F → J → D → I

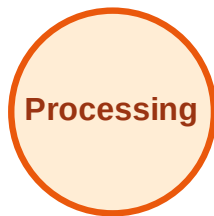
Visited: A, E, G



BFS Traversal

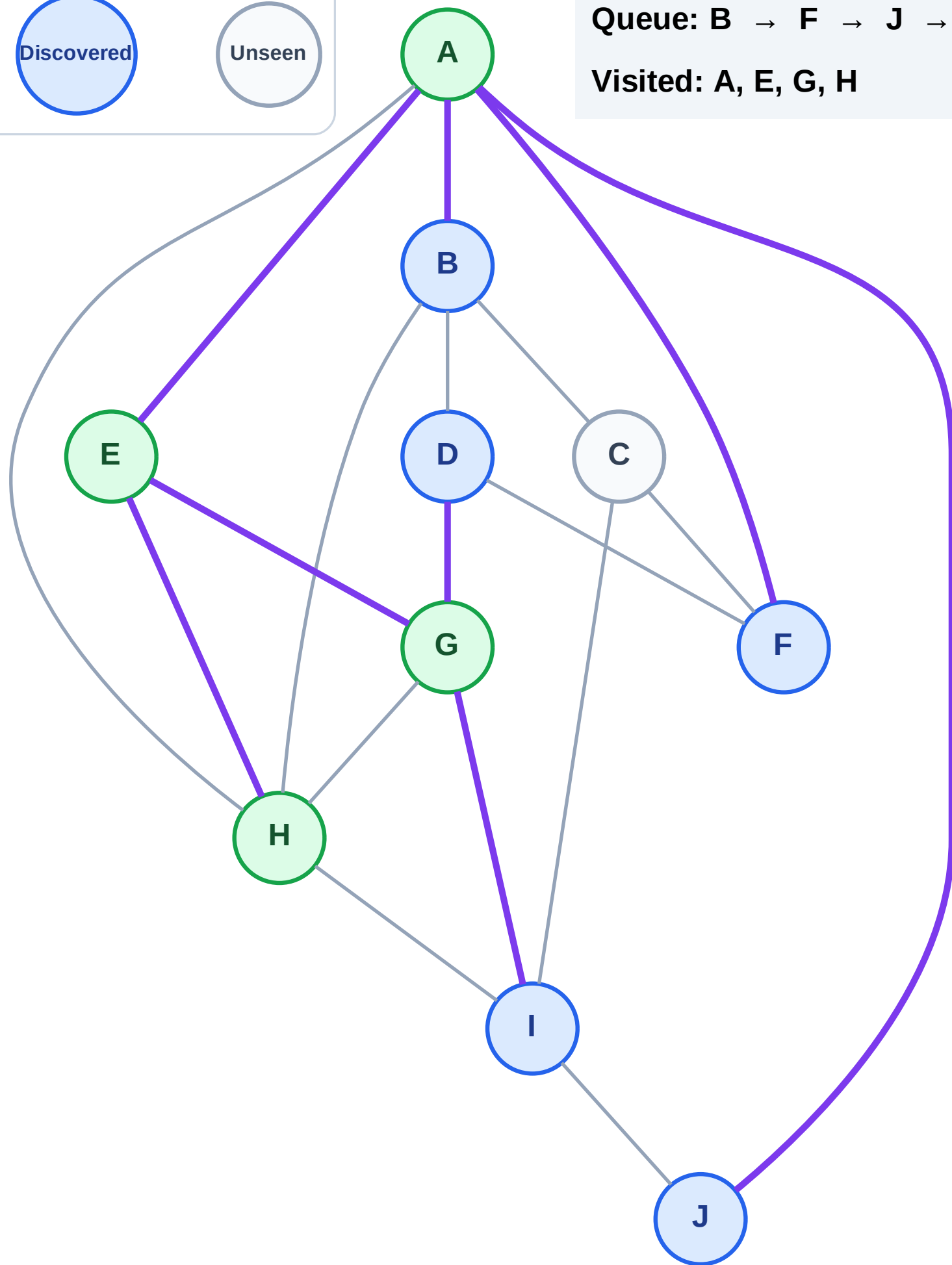
Step 9: No new neighbors from H

Legend



Queue: B → F → J → D → I

Visited: A, E, G, H



BFS Traversal

Step 10: Process node B

Legend

Complete

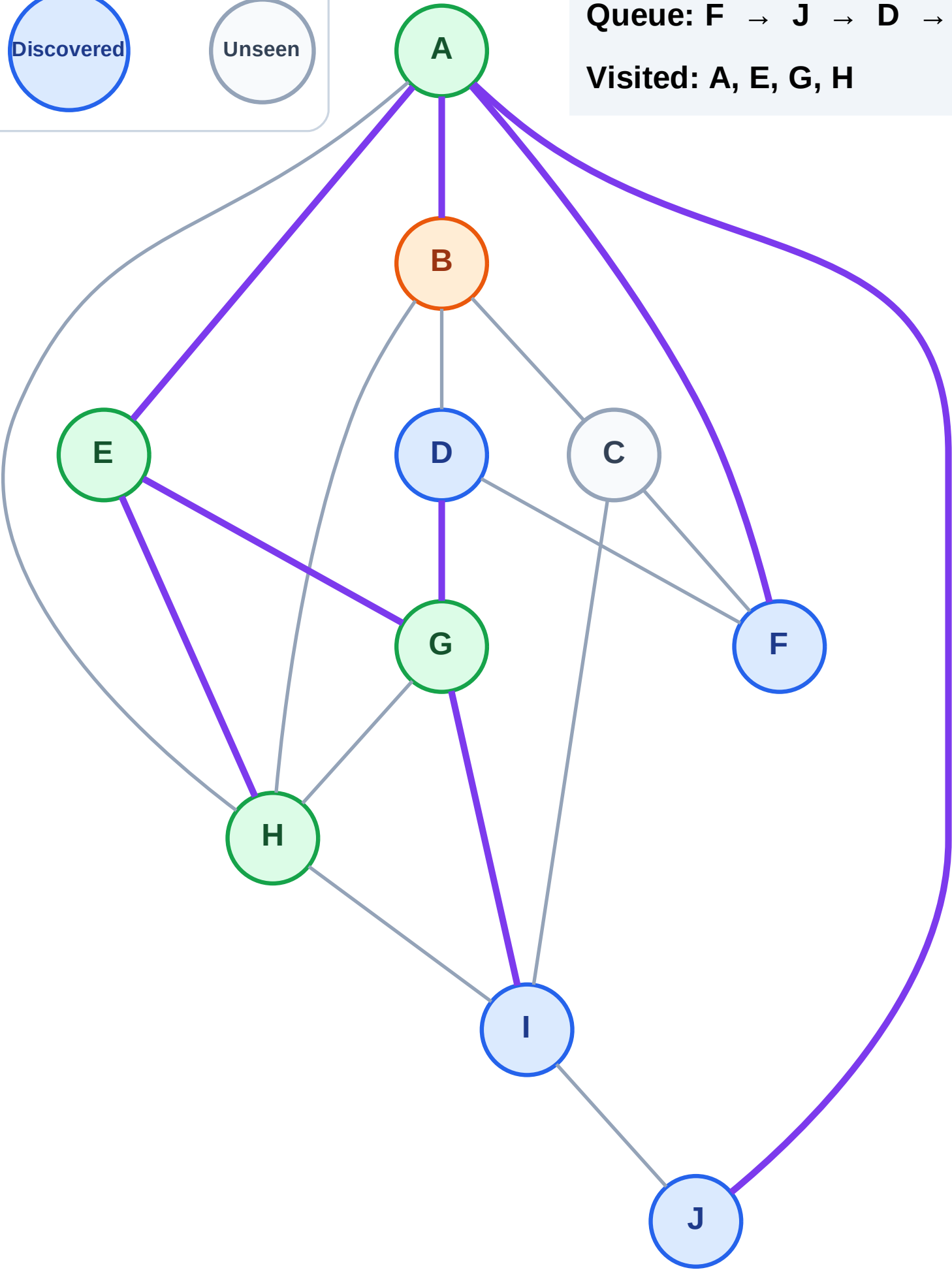
Processing

Discovered

Unseen

Queue: F → J → D → I

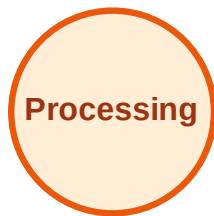
Visited: A, E, G, H



BFS Traversal

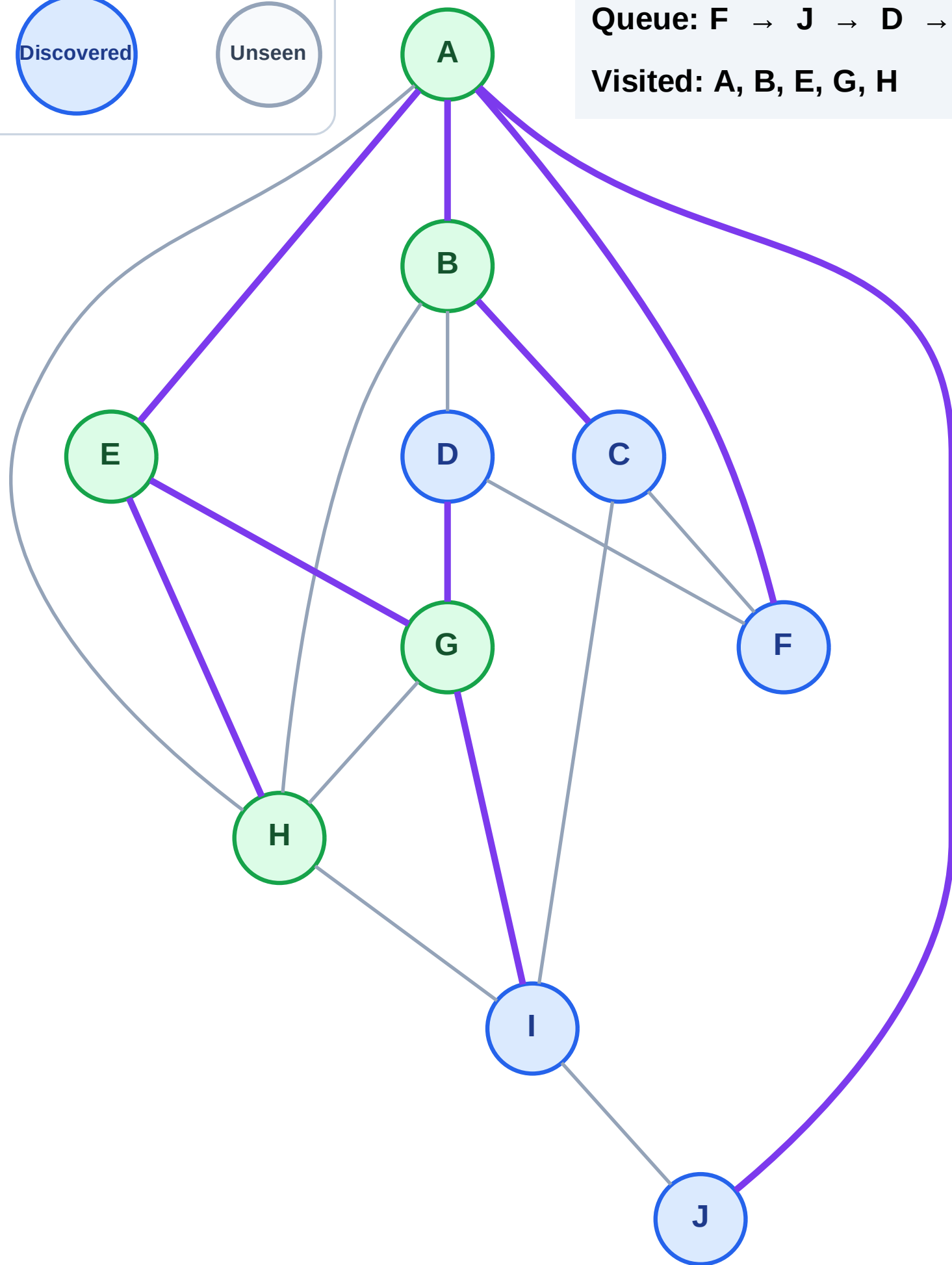
Step 11: Discover C from B

Legend



Queue: F → J → D → I → C

Visited: A, B, E, G, H



BFS Traversal

Step 12: Process node F

Legend

Complete

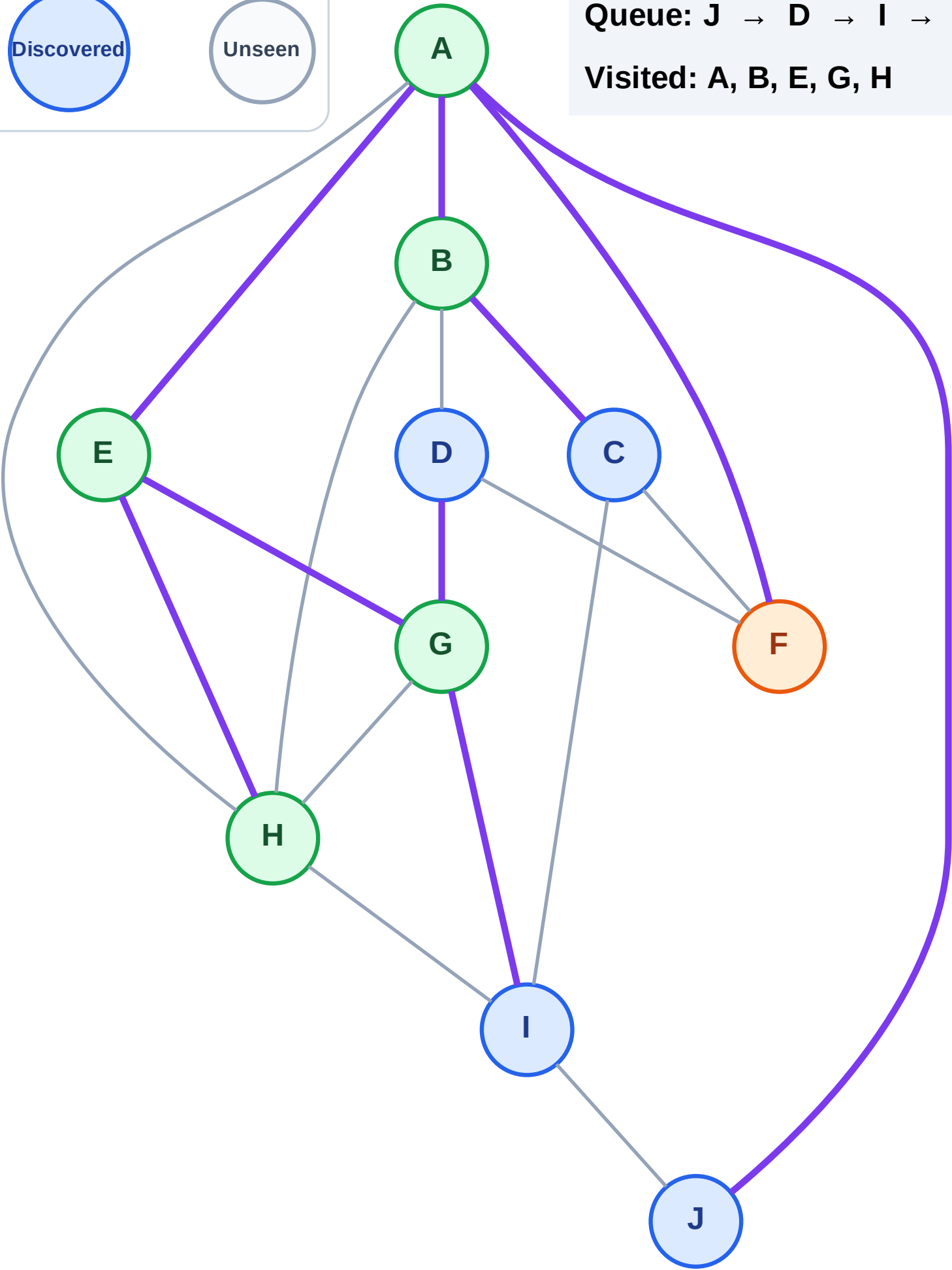
Processing

Discovered

Unseen

Queue: J → D → I → C

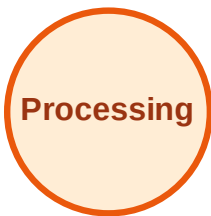
Visited: A, B, E, G, H



BFS Traversal

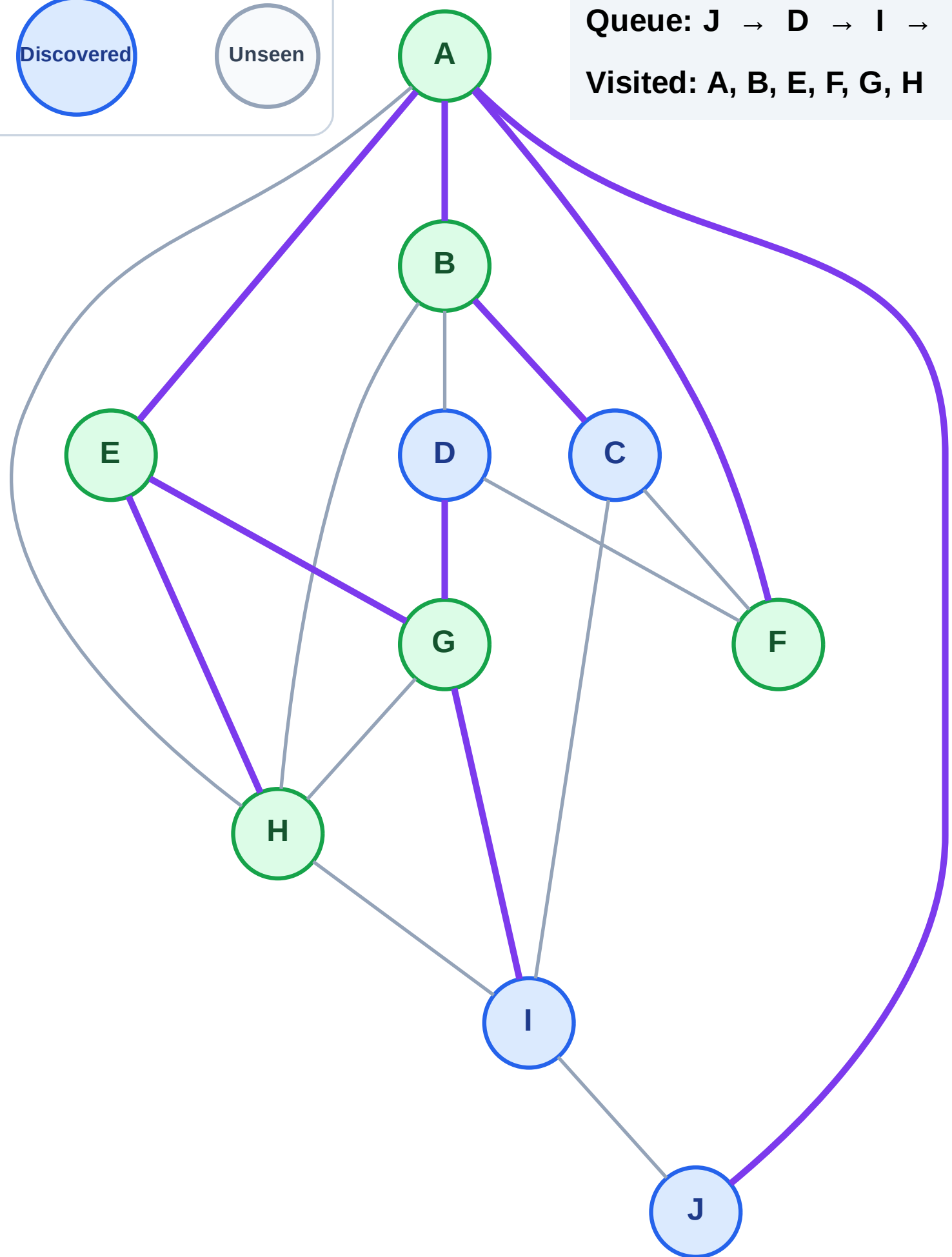
Step 13: No new neighbors from F

Legend



Queue: J → D → I → C

Visited: A, B, E, F, G, H



BFS Traversal

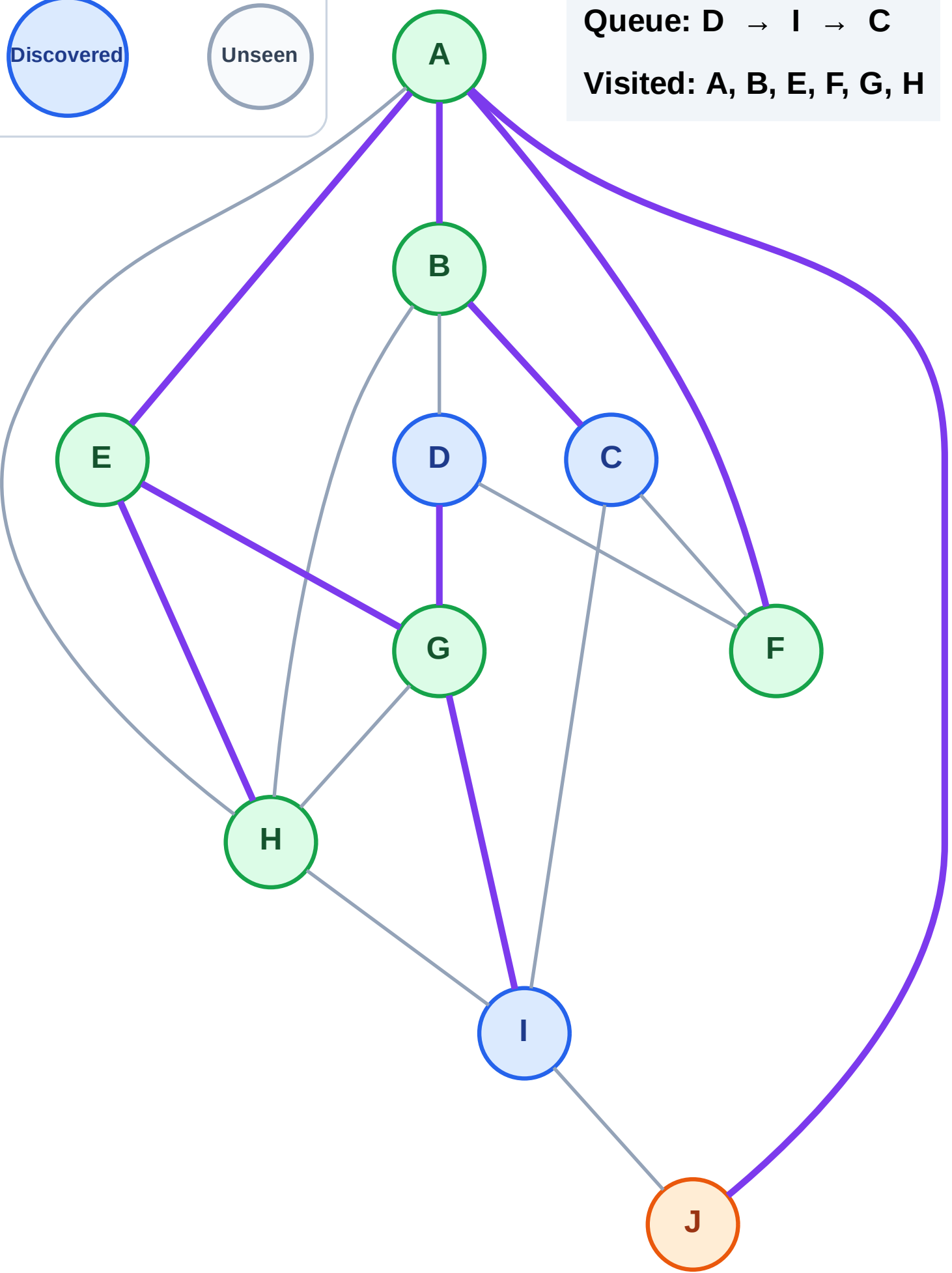
Step 14: Process node J

Legend



Queue: D → I → C

Visited: A, B, E, F, G, H



BFS Traversal

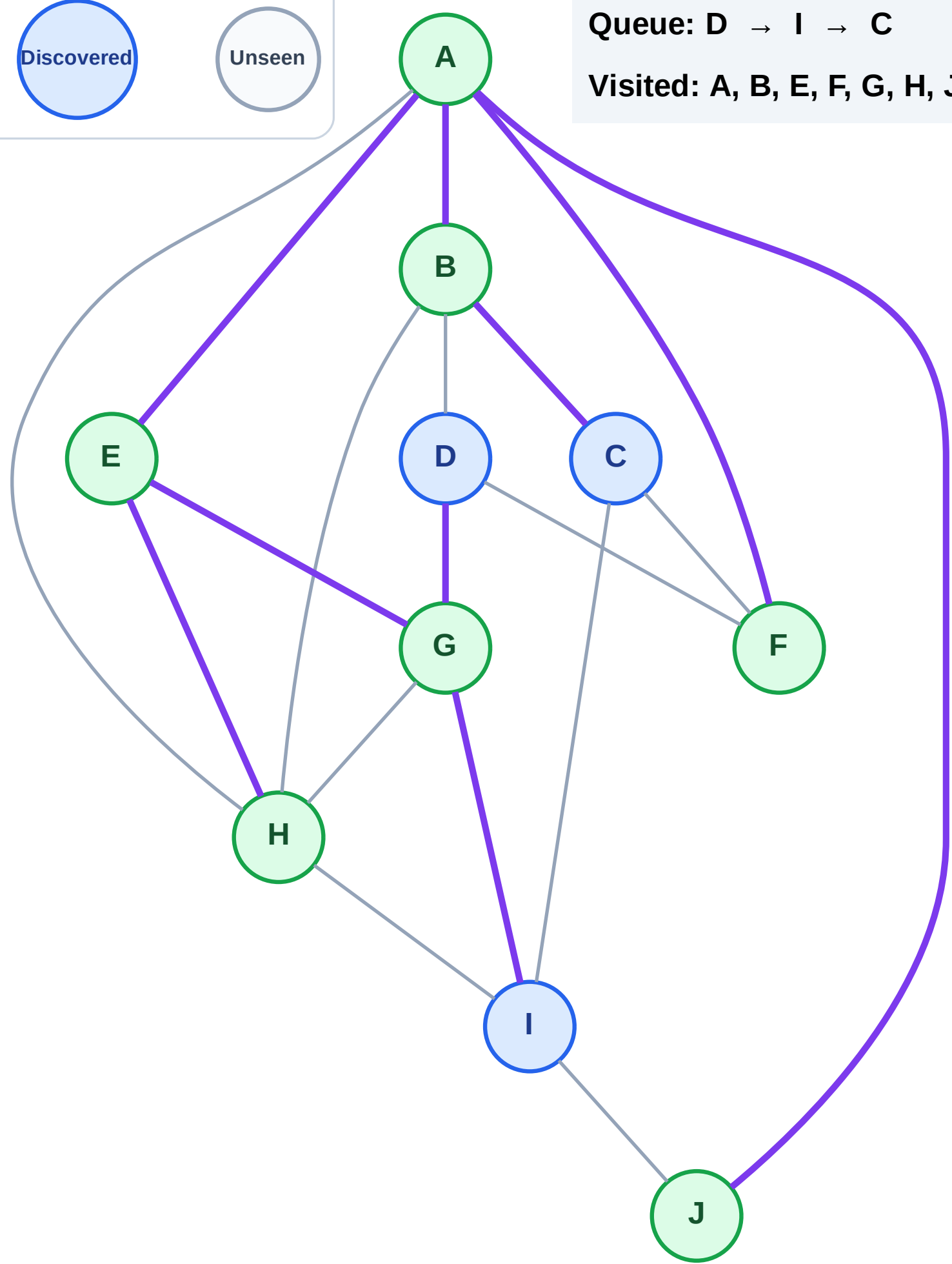
Step 15: No new neighbors from J

Legend



Queue: D → I → C

Visited: A, B, E, F, G, H, J



BFS Traversal

Step 16: Process node D

Legend

Complete

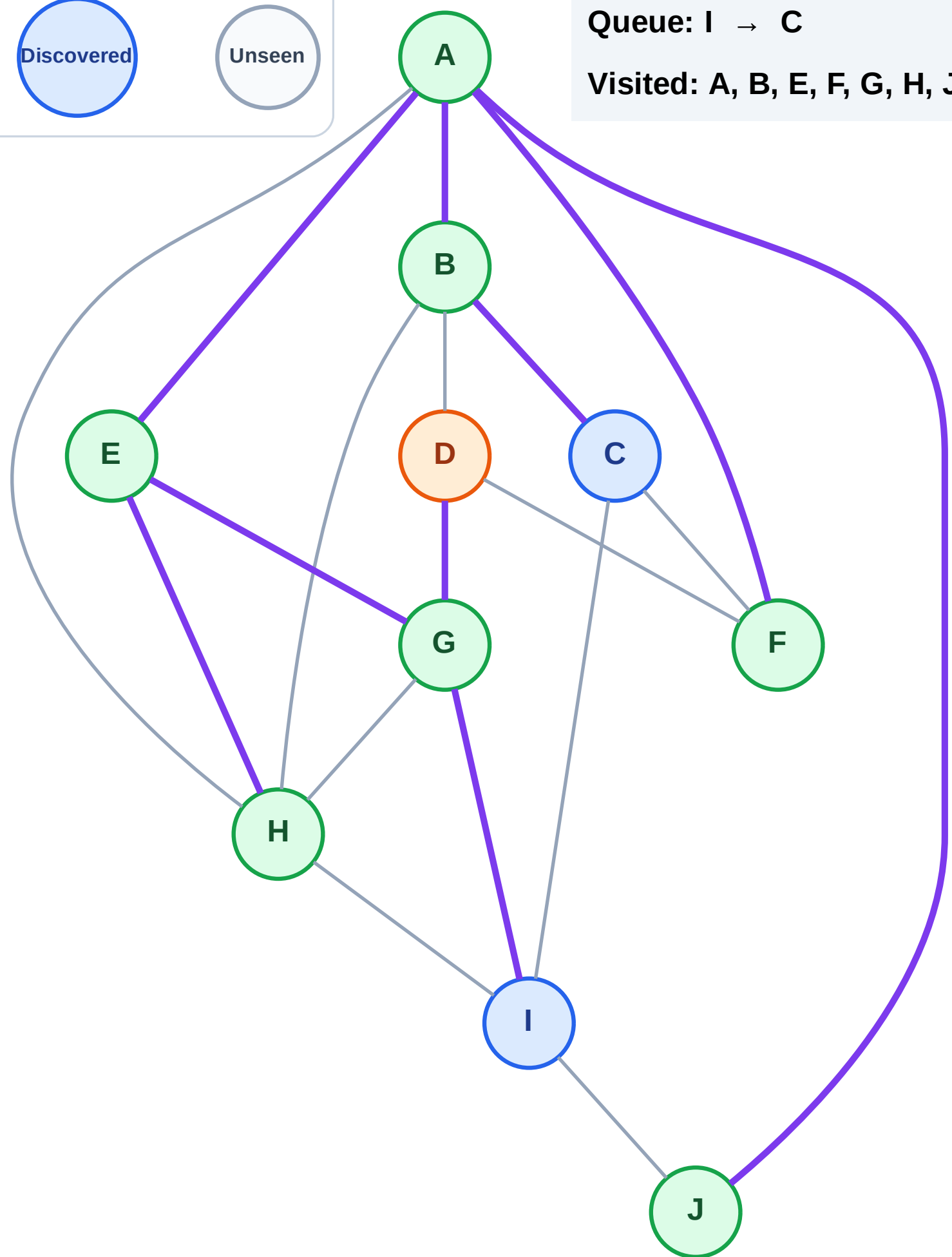
Processing

Discovered

Unseen

Queue: I → C

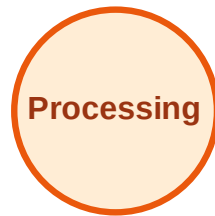
Visited: A, B, E, F, G, H, J



BFS Traversal

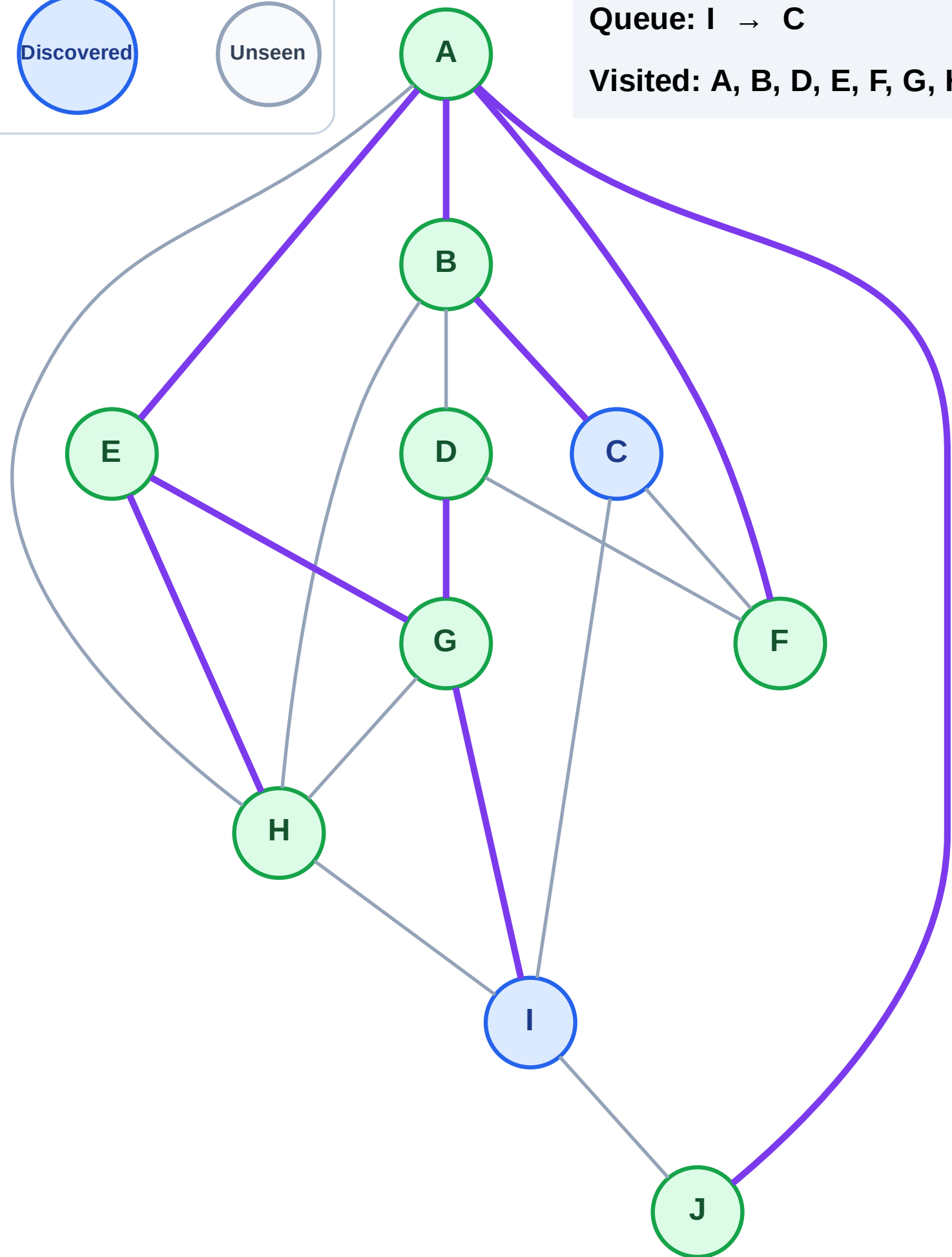
Step 17: No new neighbors from D

Legend



Queue: I → C

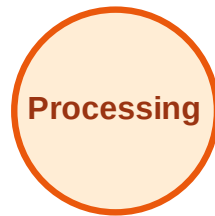
Visited: A, B, D, E, F, G, H, J



BFS Traversal

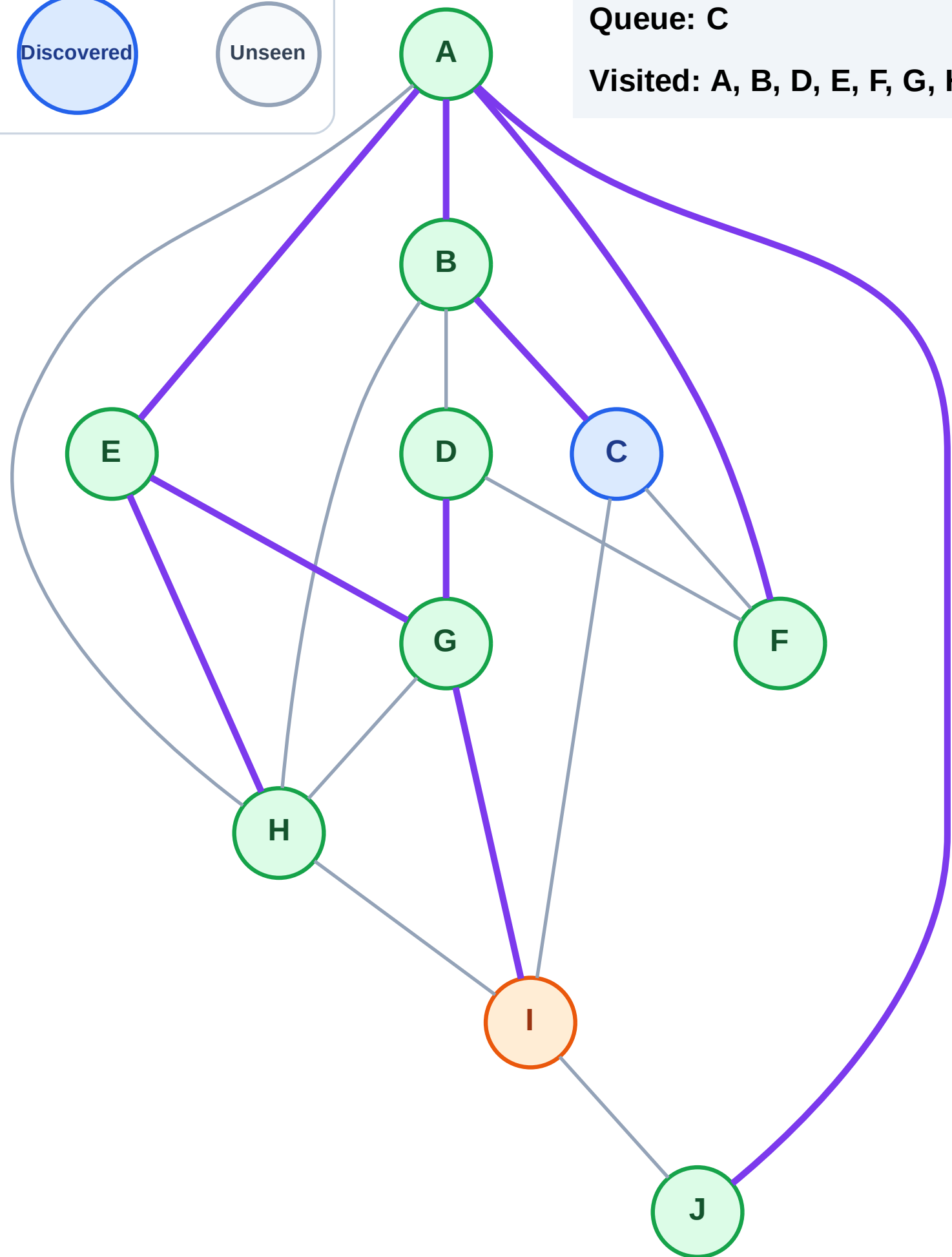
Step 18: Process node I

Legend



Queue: C

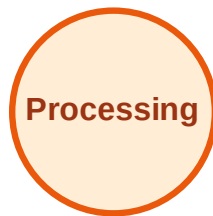
Visited: A, B, D, E, F, G, H, J



BFS Traversal

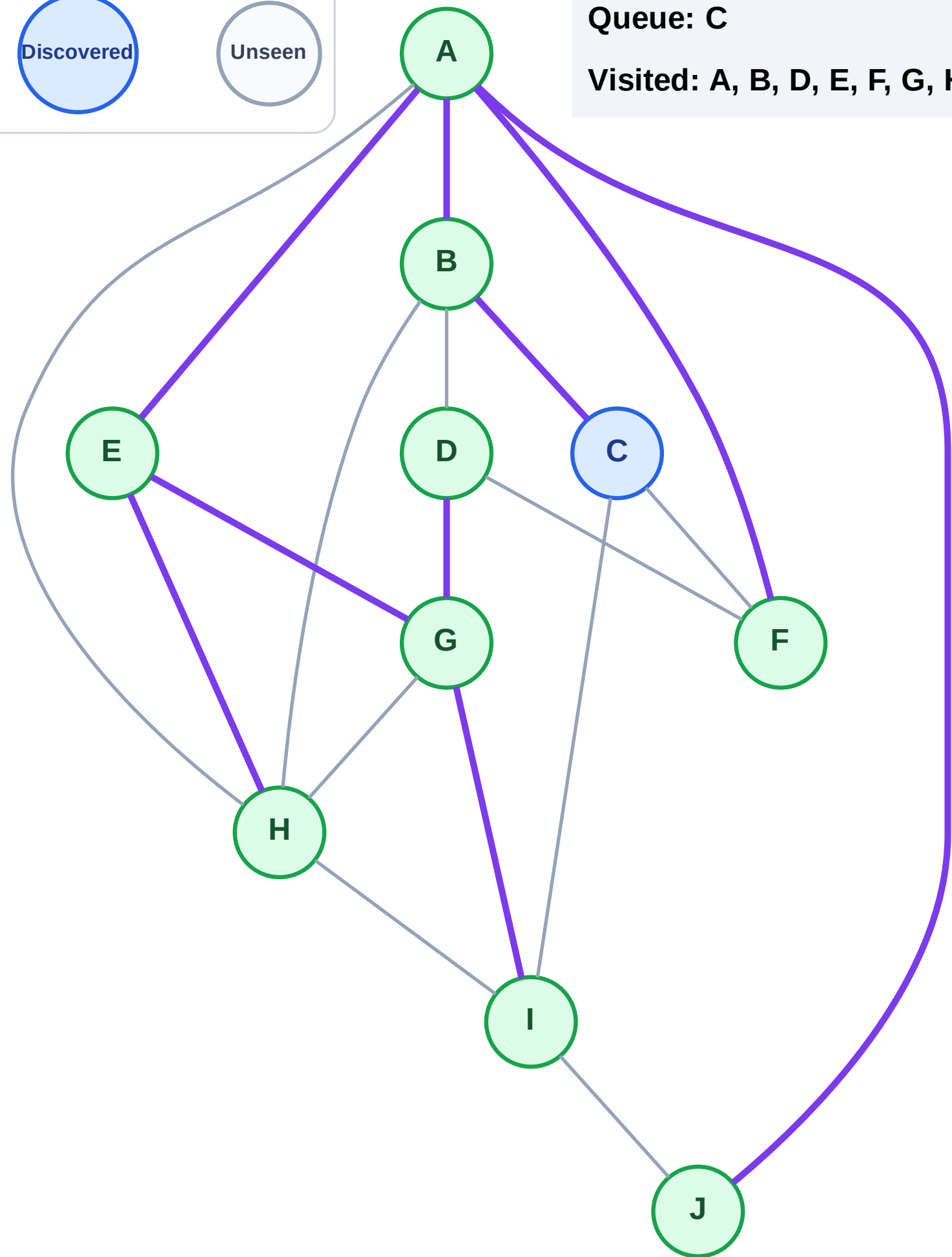
Step 19: No new neighbors from I

Legend



Queue: C

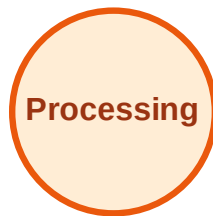
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

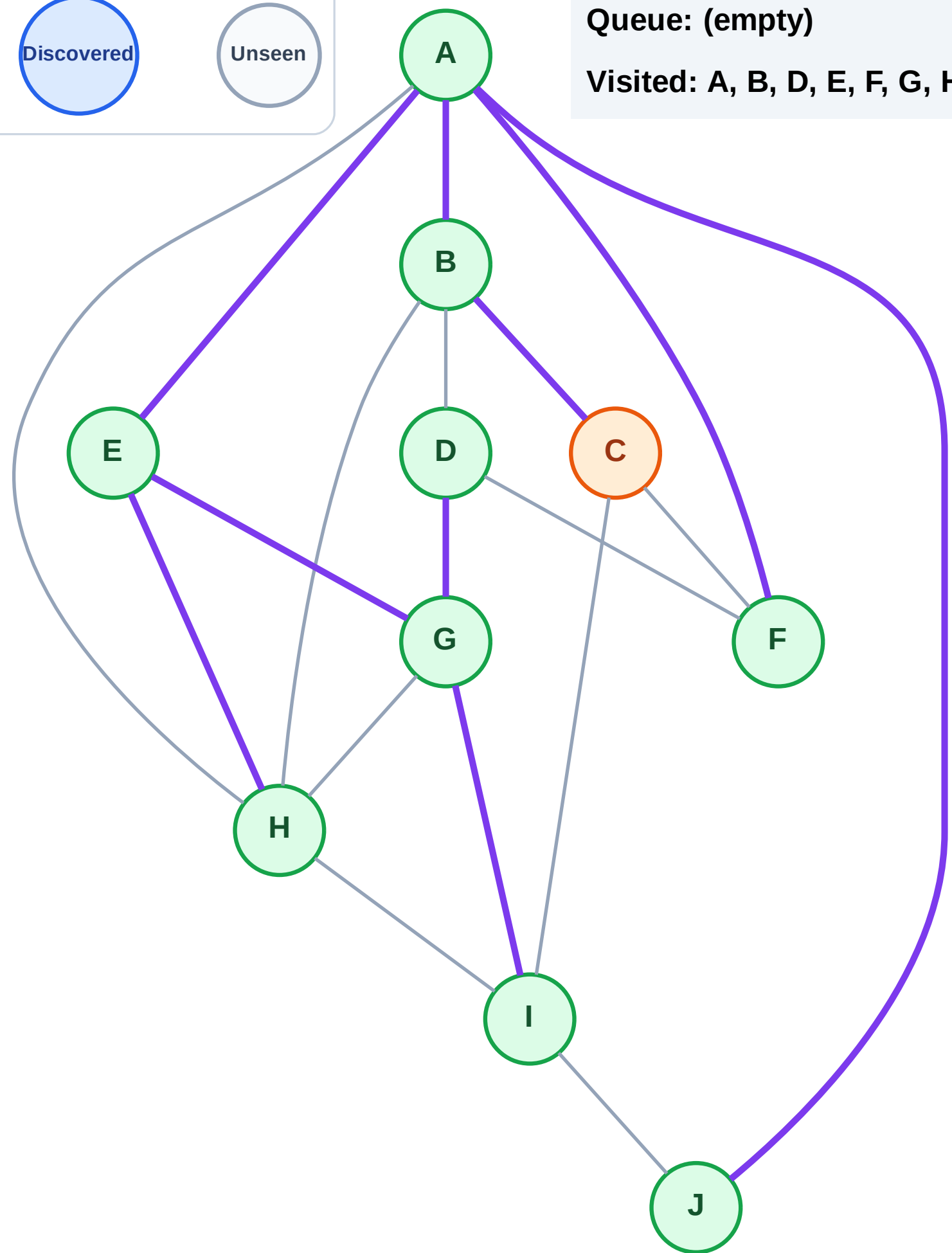
Step 20: Process node C

Legend



Queue: (empty)

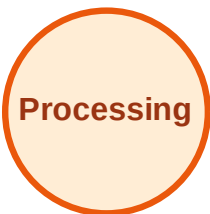
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

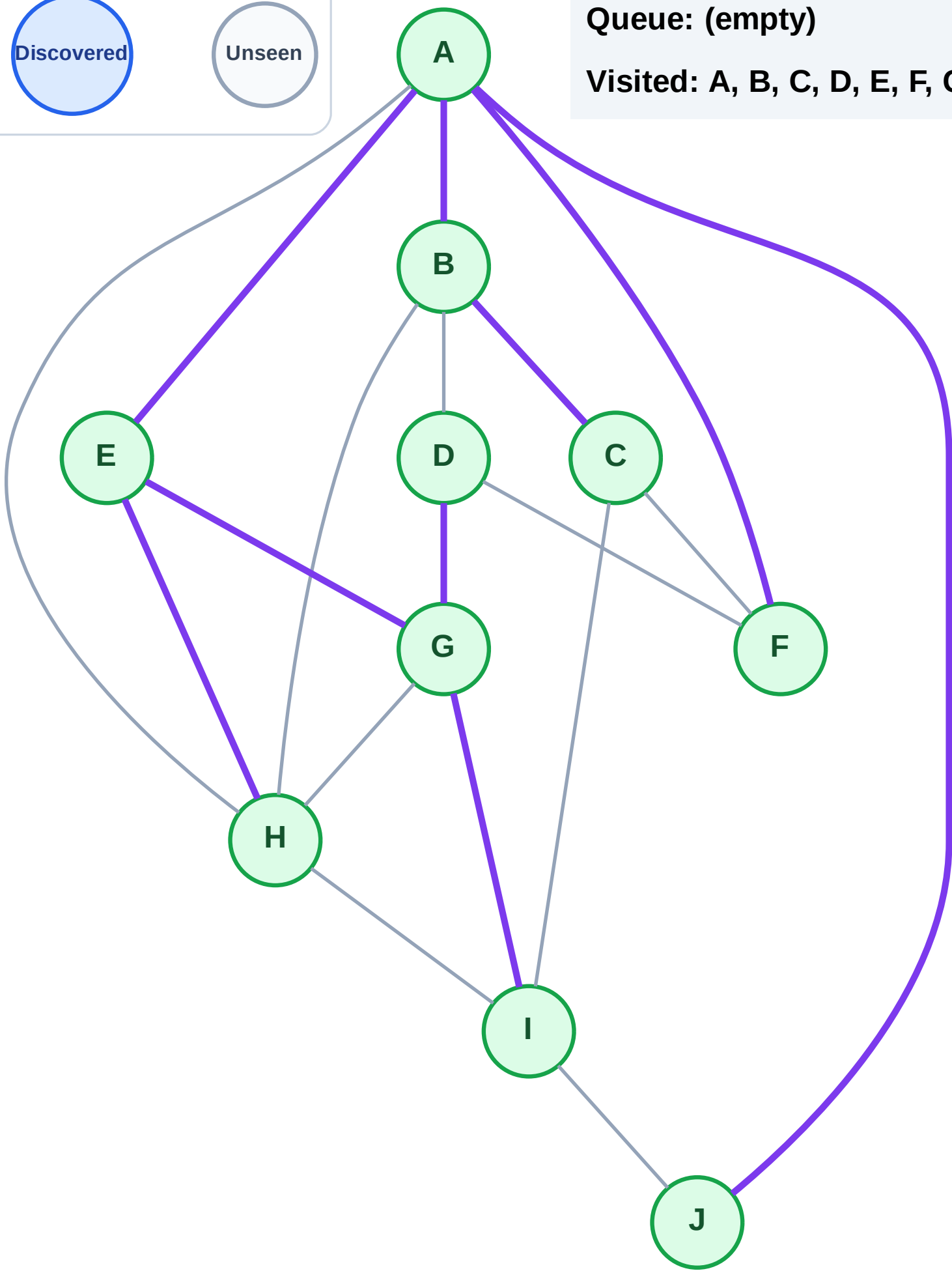
Step 21: No new neighbors from C

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

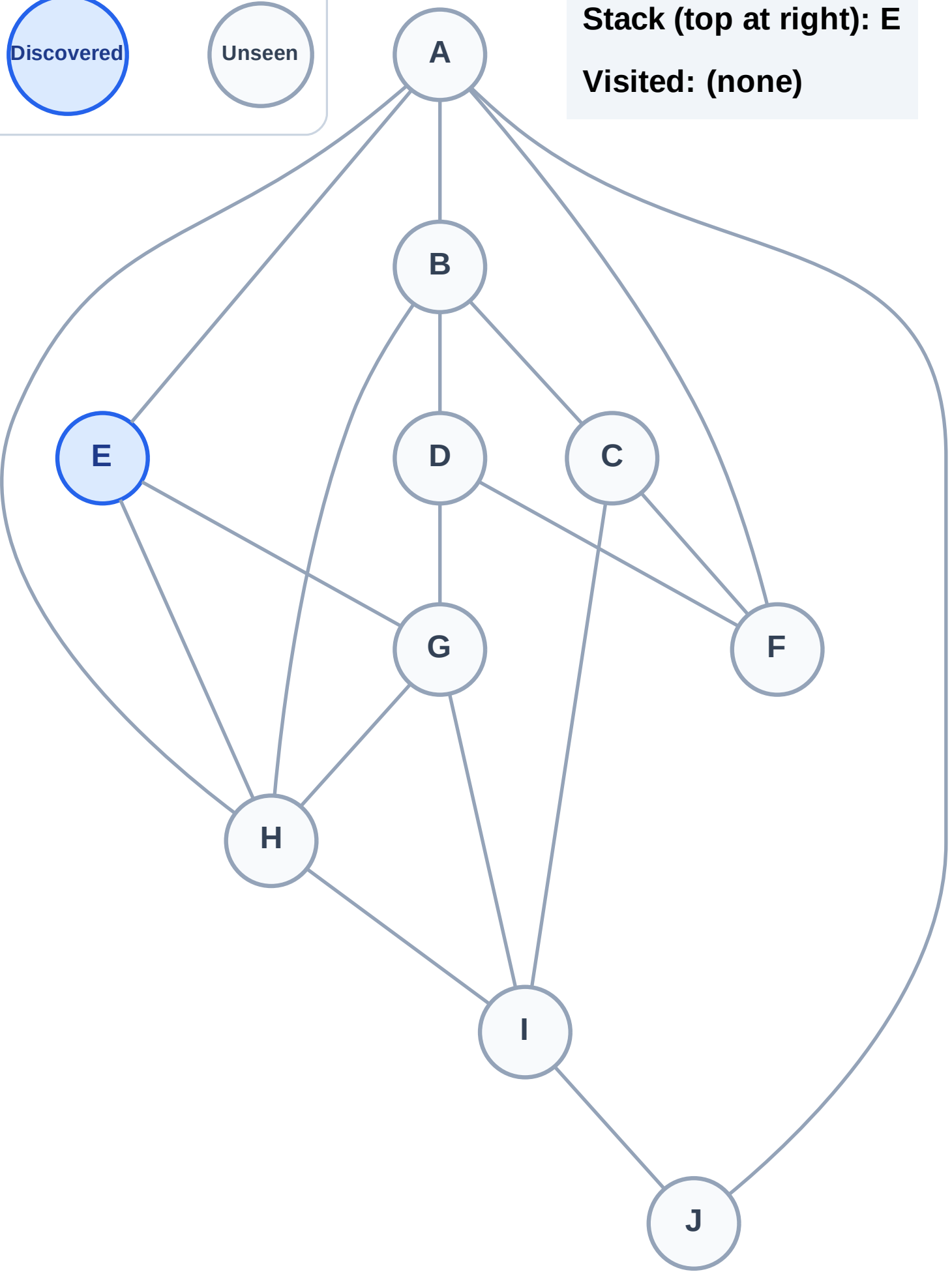
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

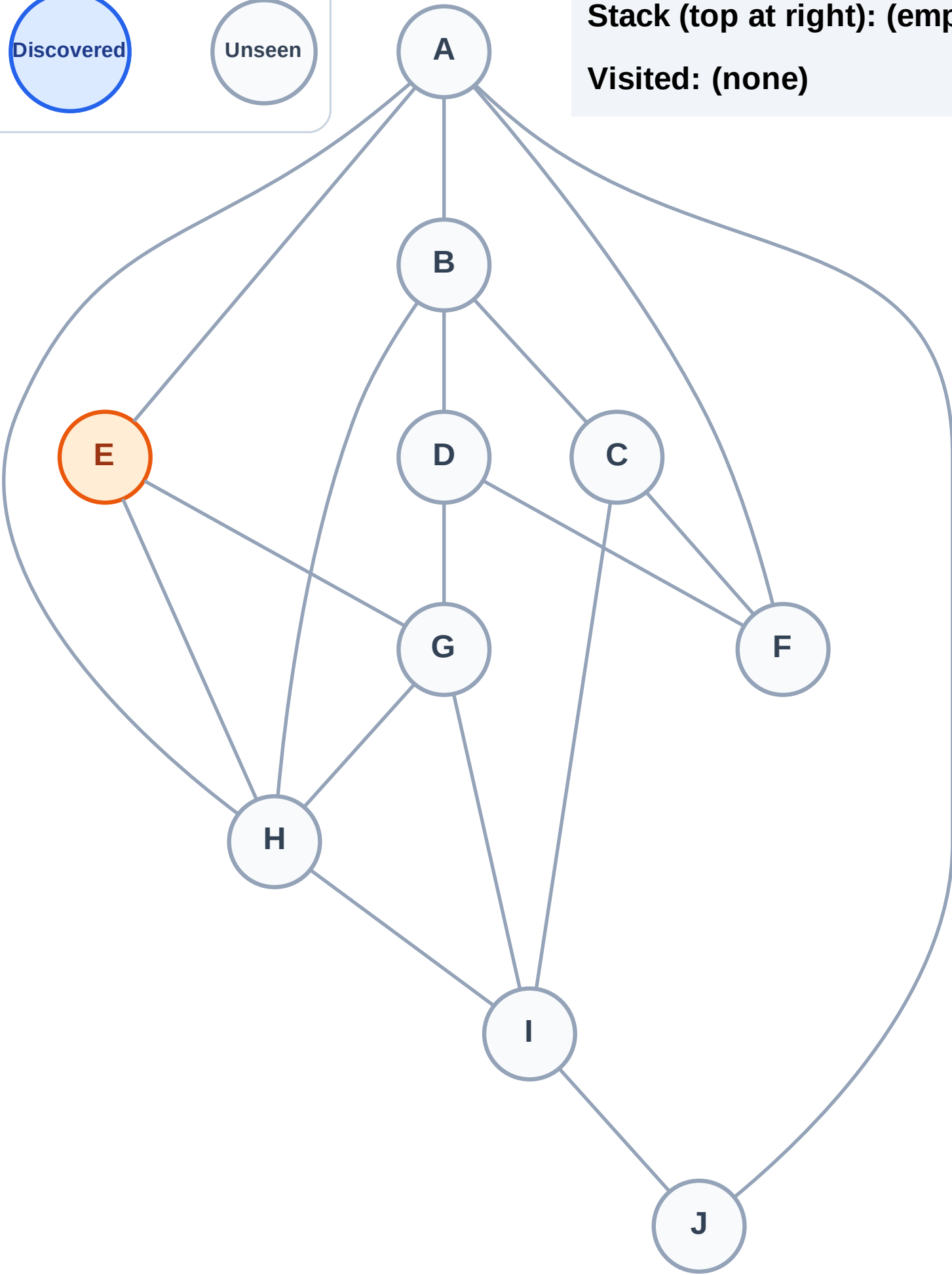
Complete

Processing

Discovered

Unseen

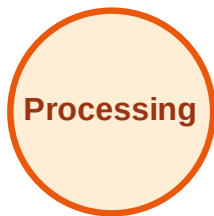
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

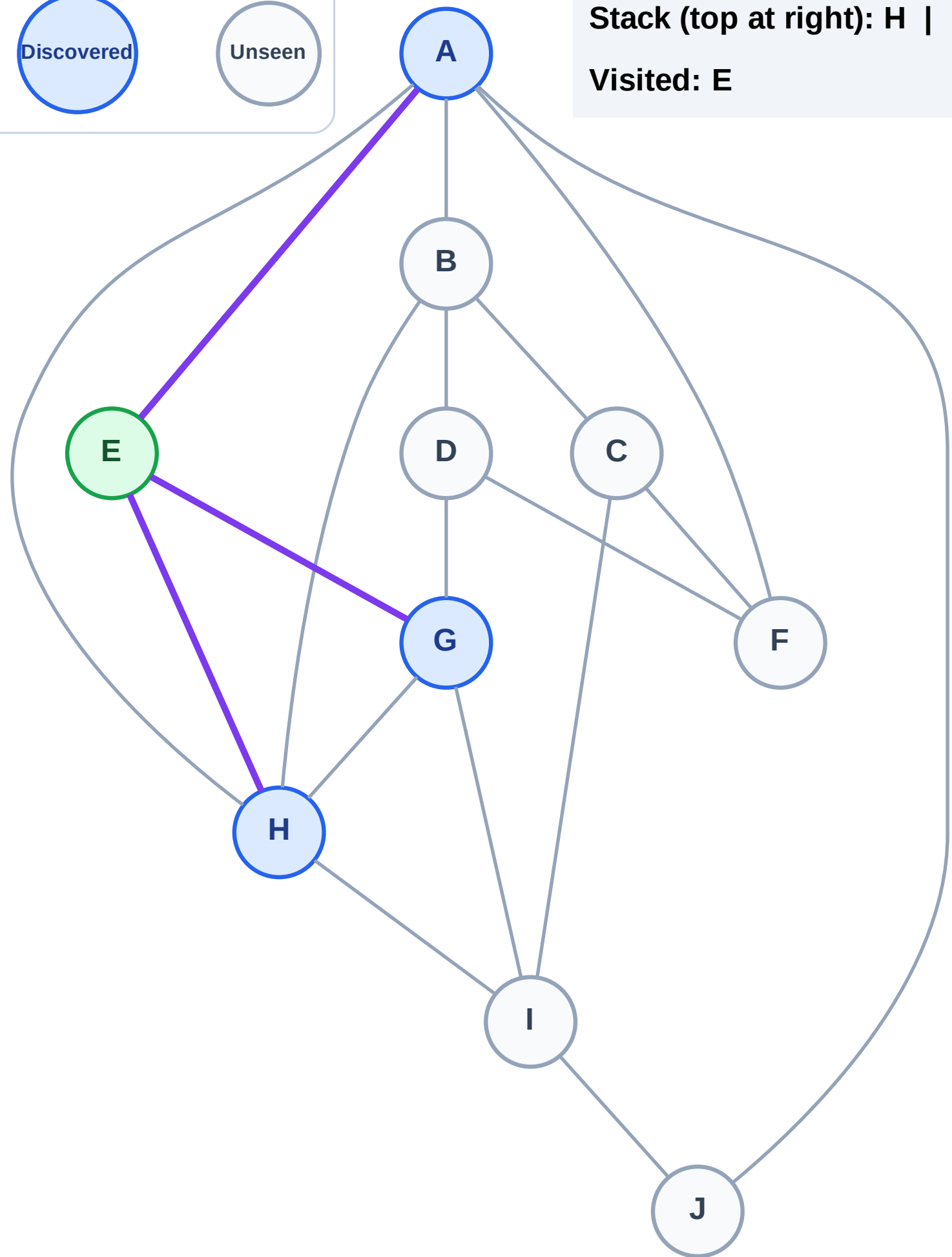
Step 3: Discover H, G, A from E

Legend



Stack (top at right): H | G | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

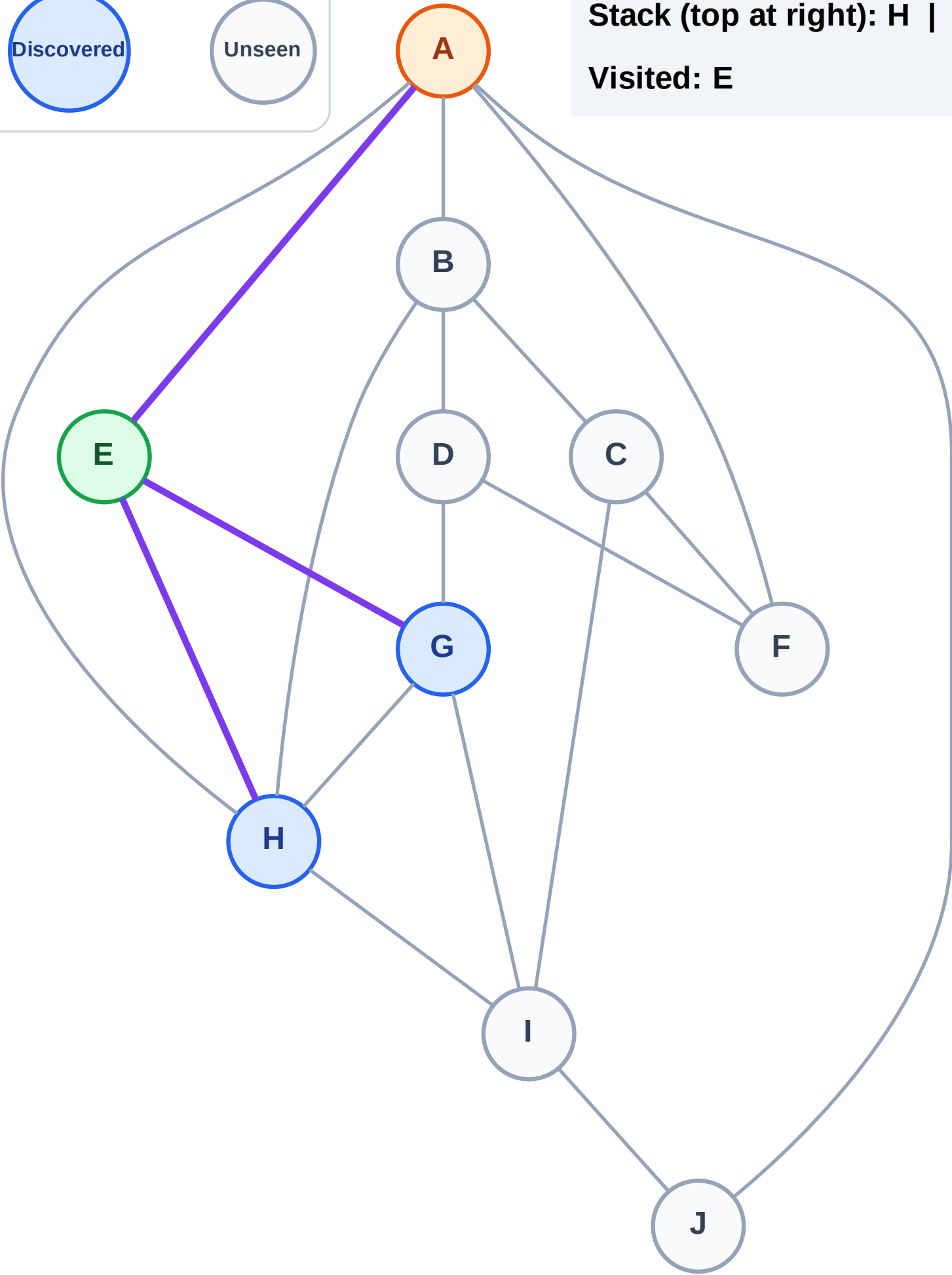
Processing

Discovered

Unseen

Stack (top at right): H | G

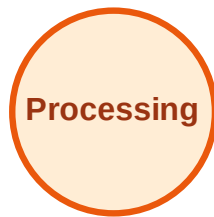
Visited: E



DFS Traversal

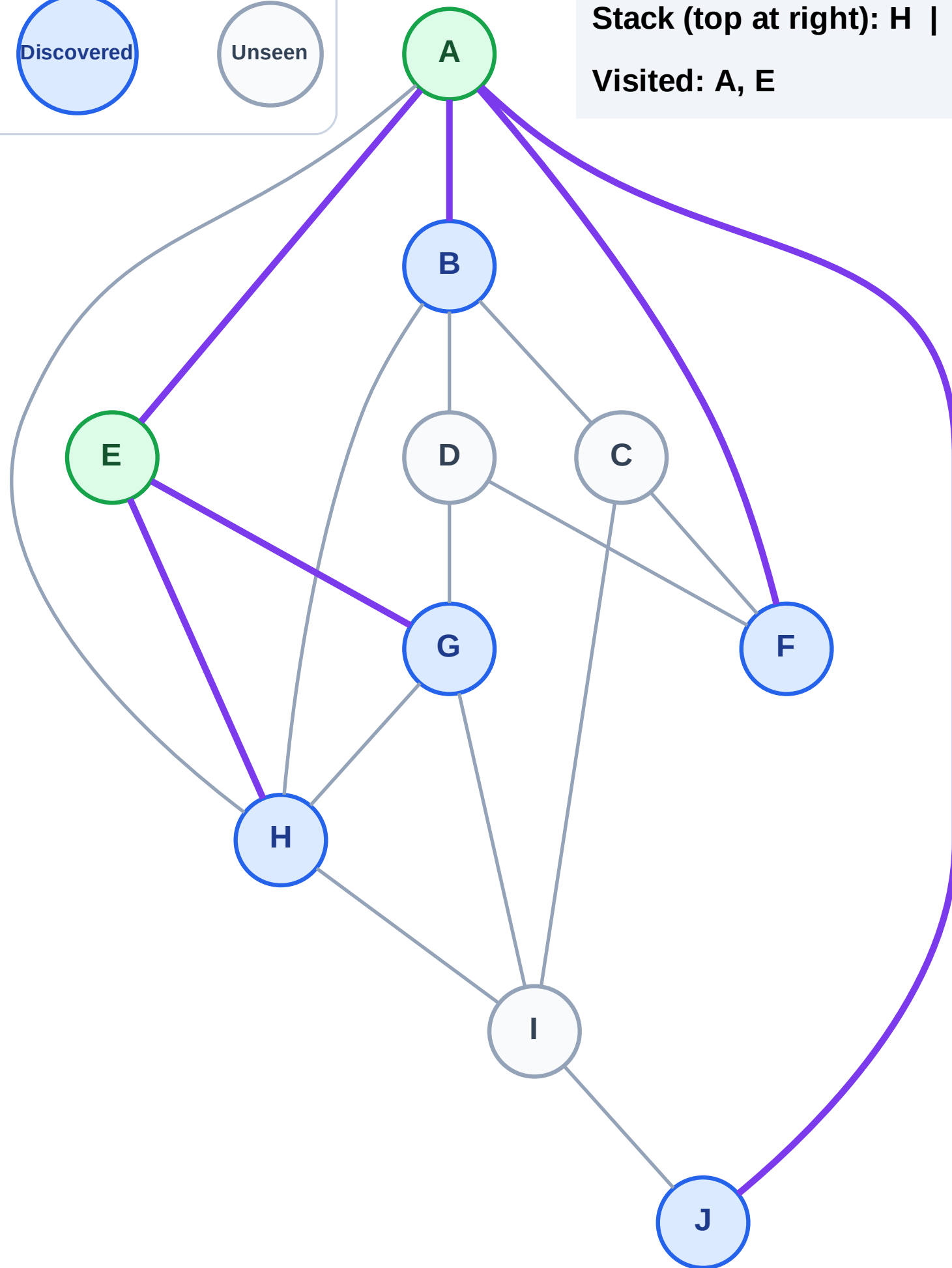
Step 5: Discover J, F, B from A

Legend



Stack (top at right): H | G | J | F | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

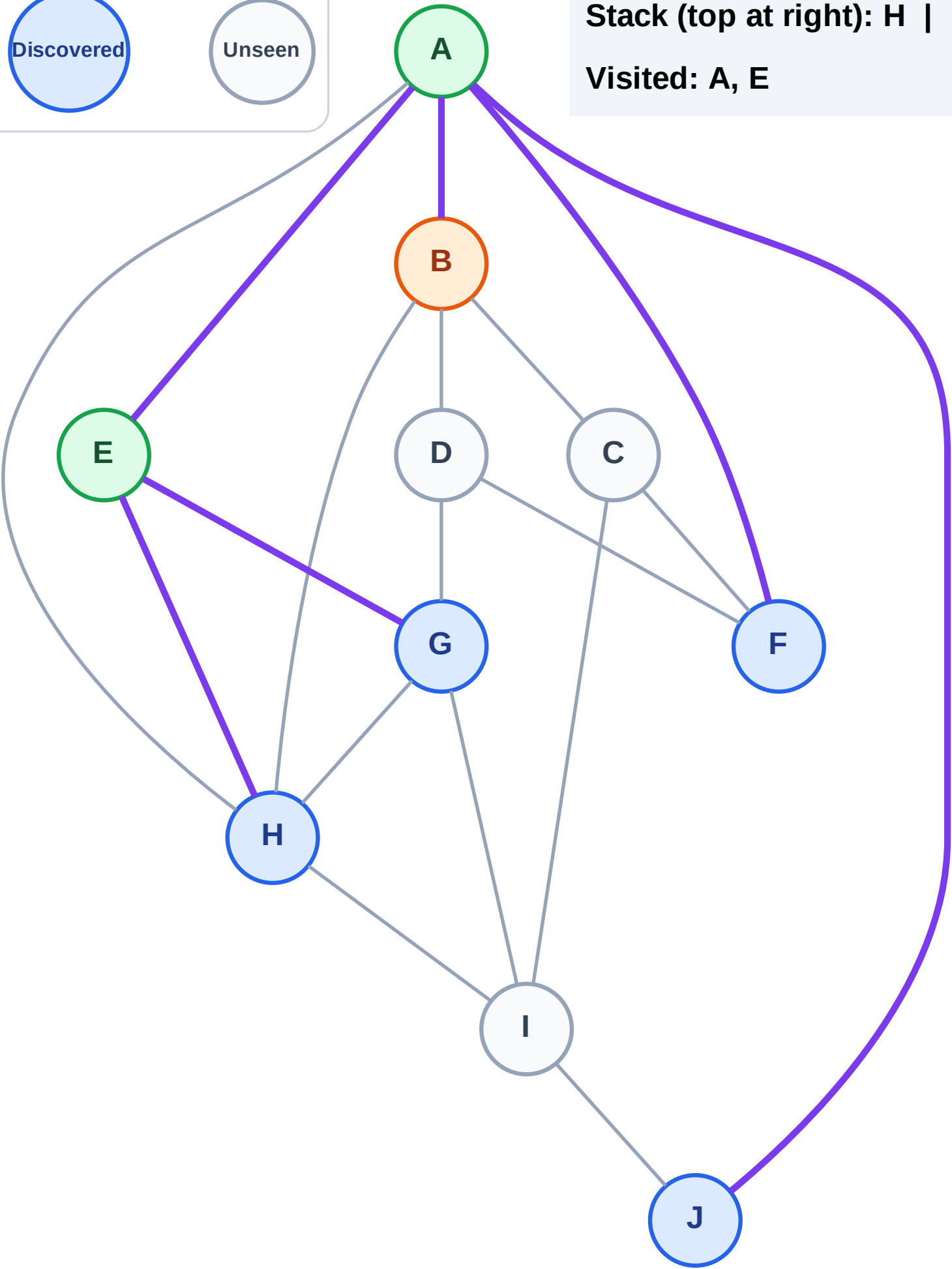
Complete

Processing

Discovered

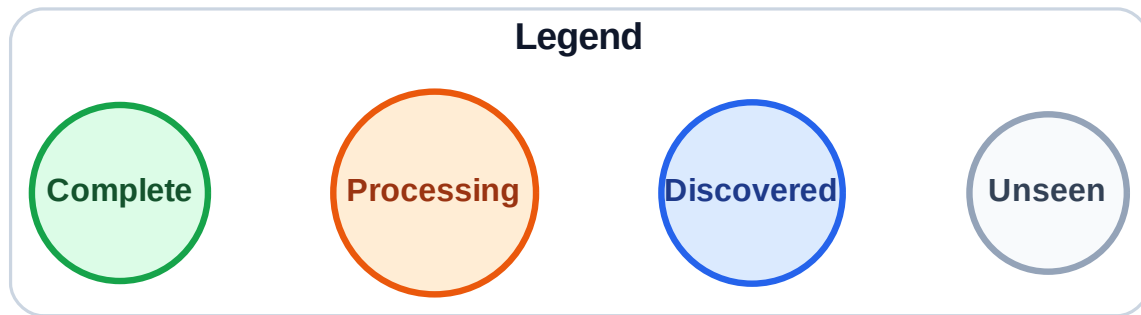
Unseen

Stack (top at right): H | G | J | F
Visited: A, E



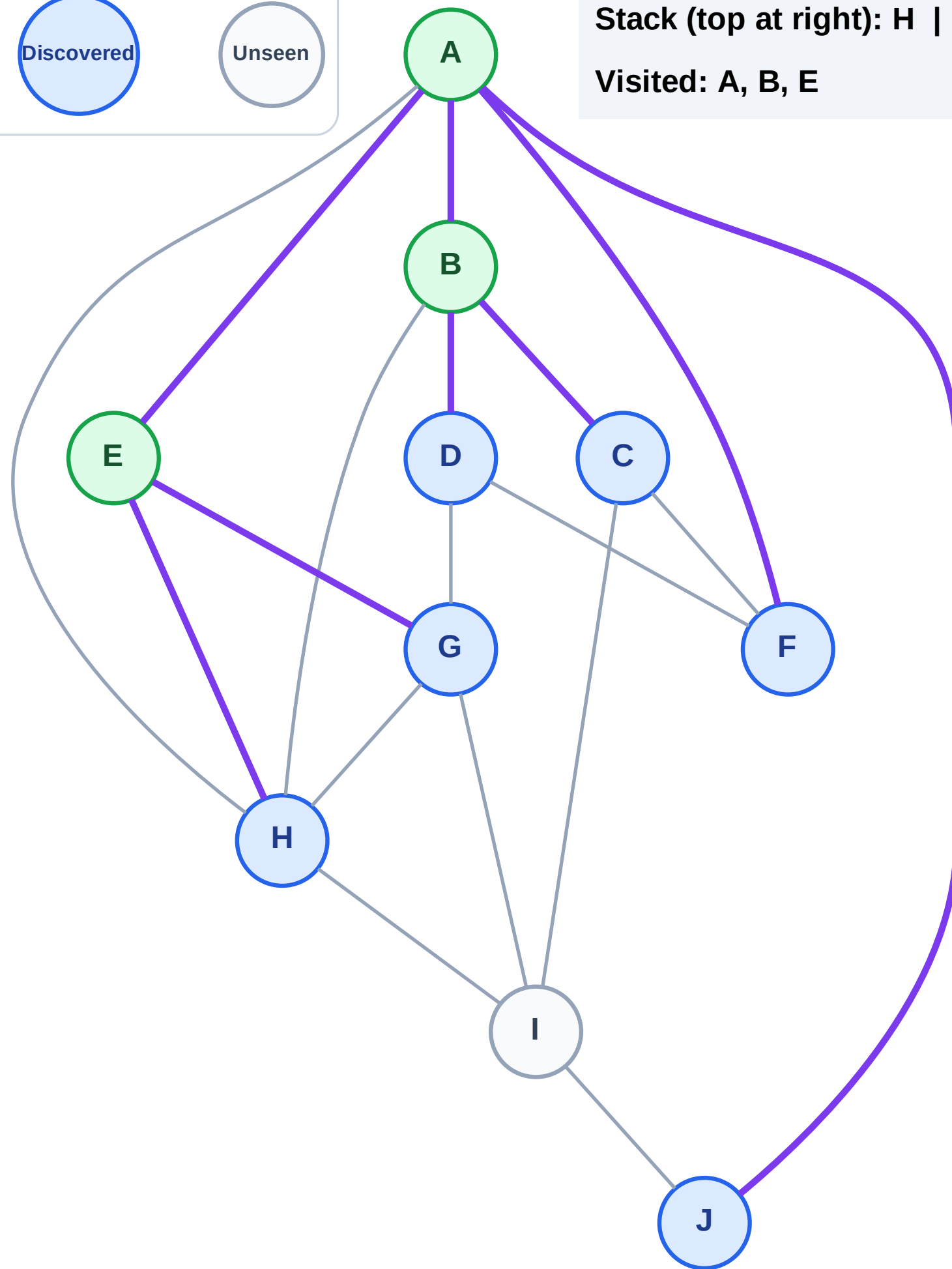
Step 7: Discover D, C from B

Step 7: Discover D, C from B



Stack (top at right): H | G | J | F | D | C

Visited: A, B, E



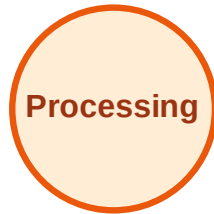
DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



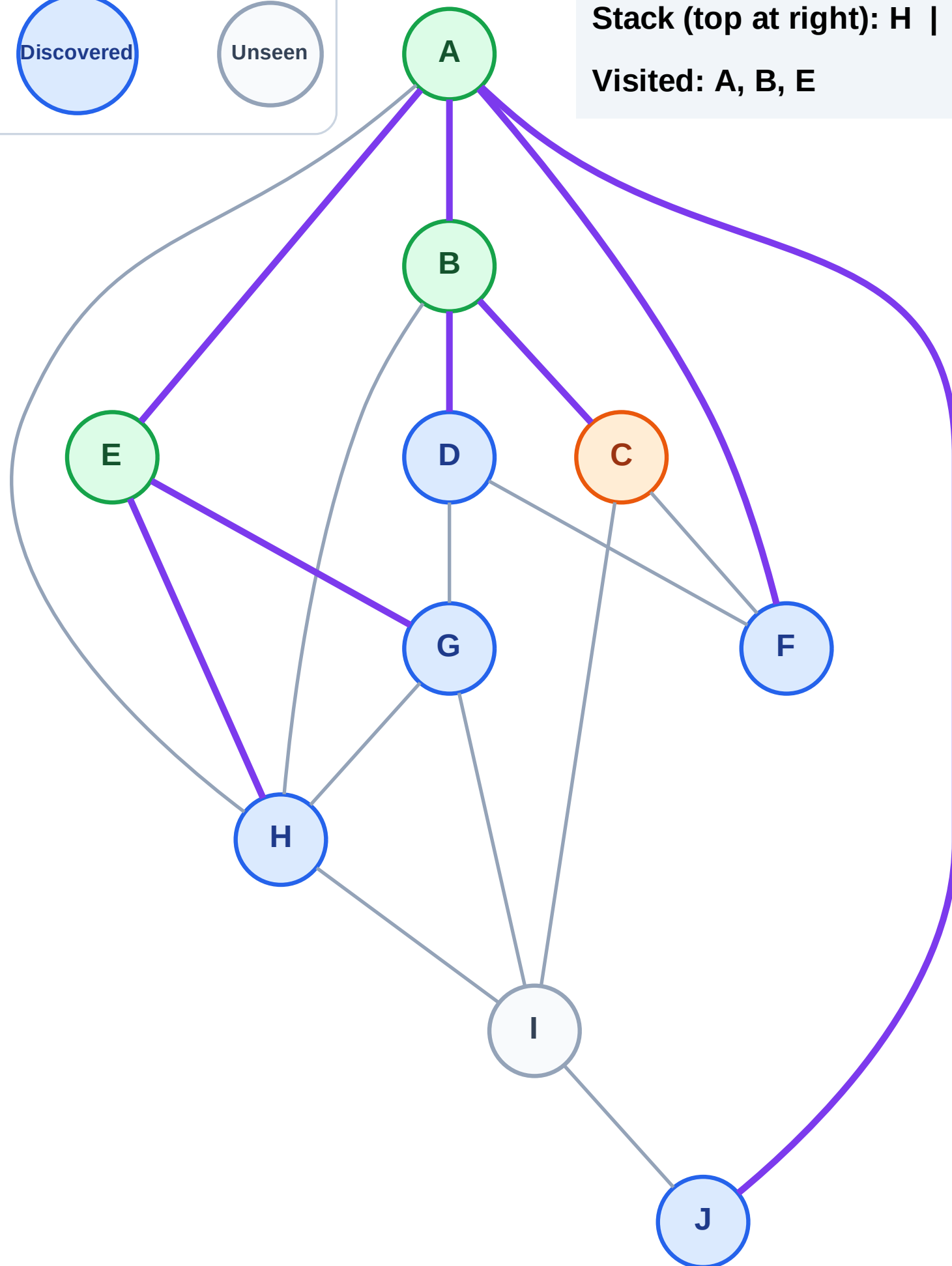
Discovered



Unseen

Stack (top at right): H | G | J | F | D

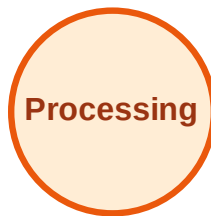
Visited: A, B, E



DFS Traversal

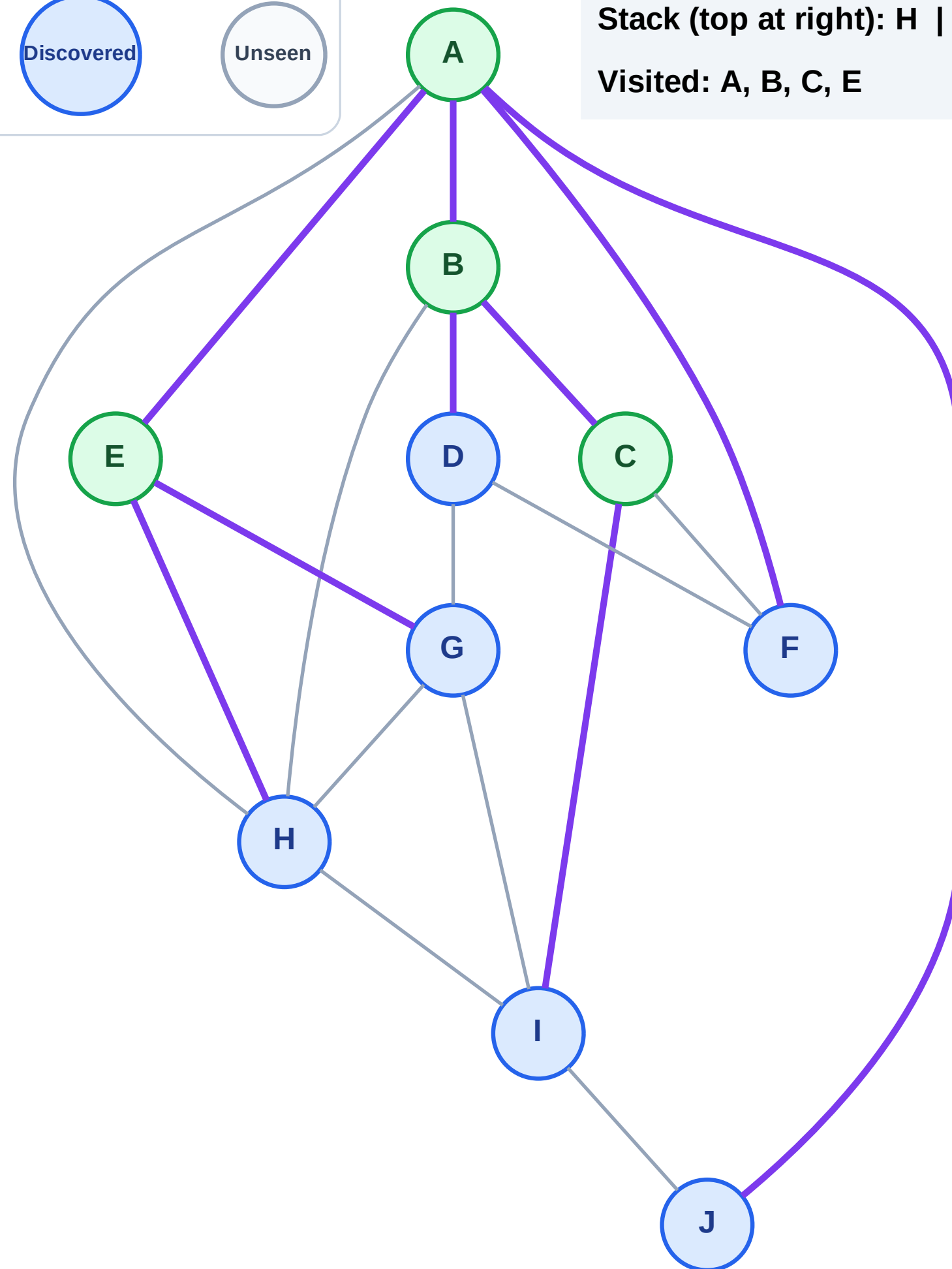
Step 9: Discover I from C

Legend



Stack (top at right): H | G | J | F | D | I

Visited: A, B, C, E



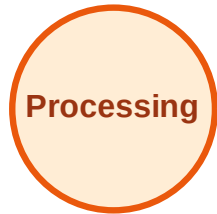
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



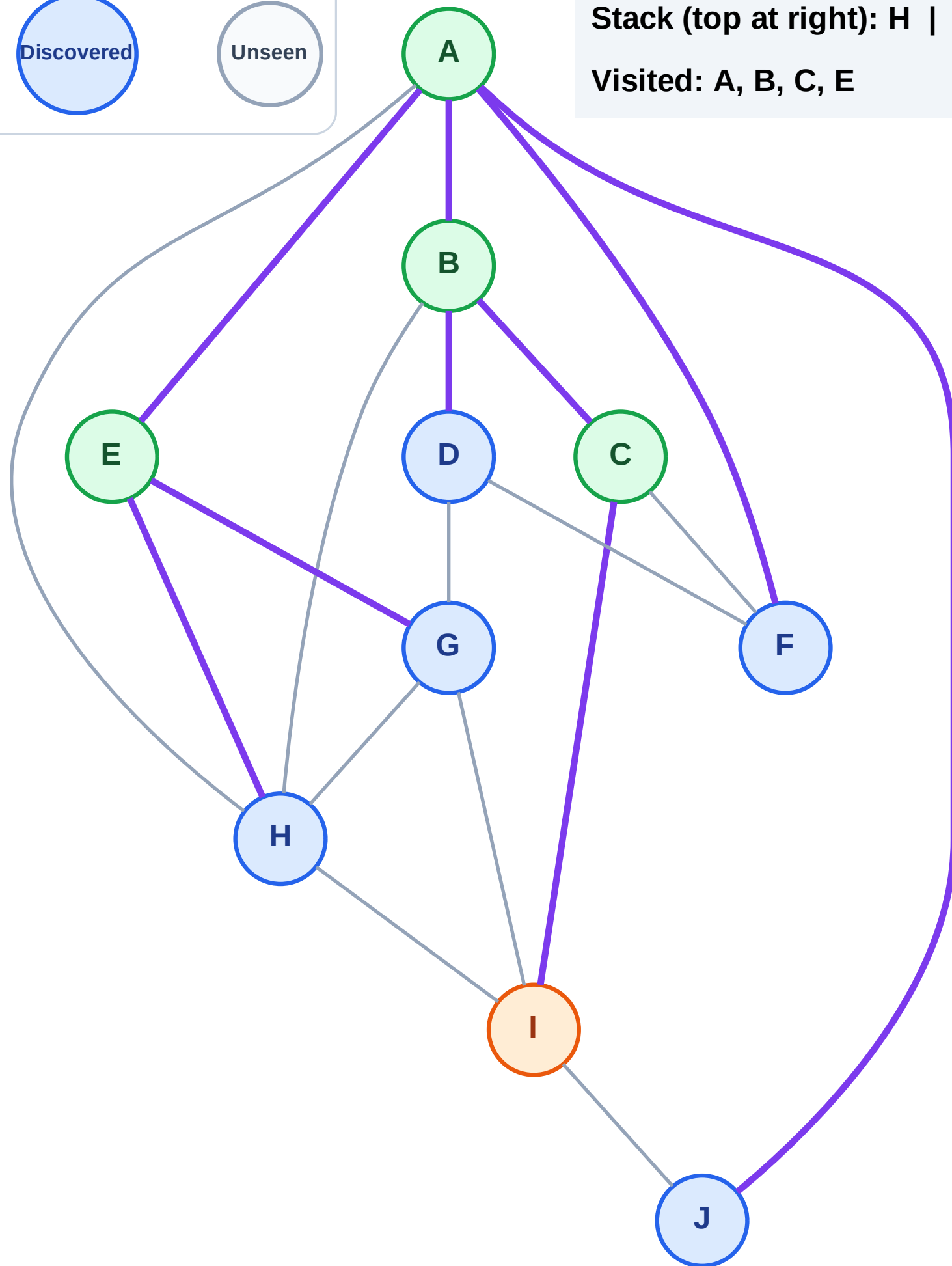
Discovered



Unseen

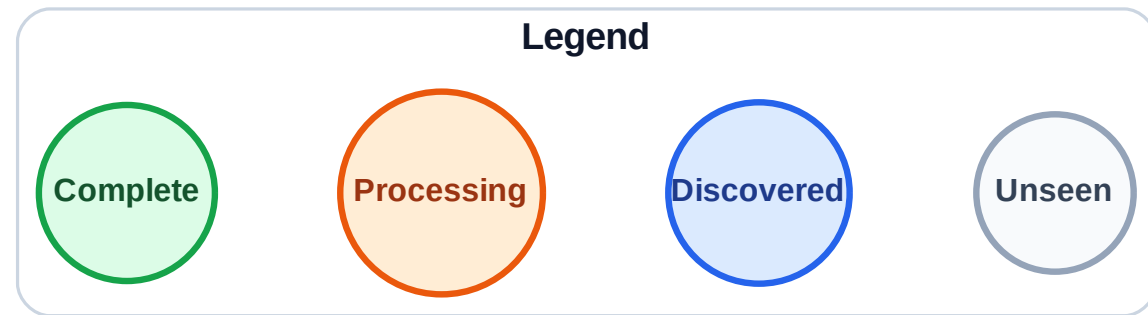
Stack (top at right): H | G | J | F | D

Visited: A, B, C, E



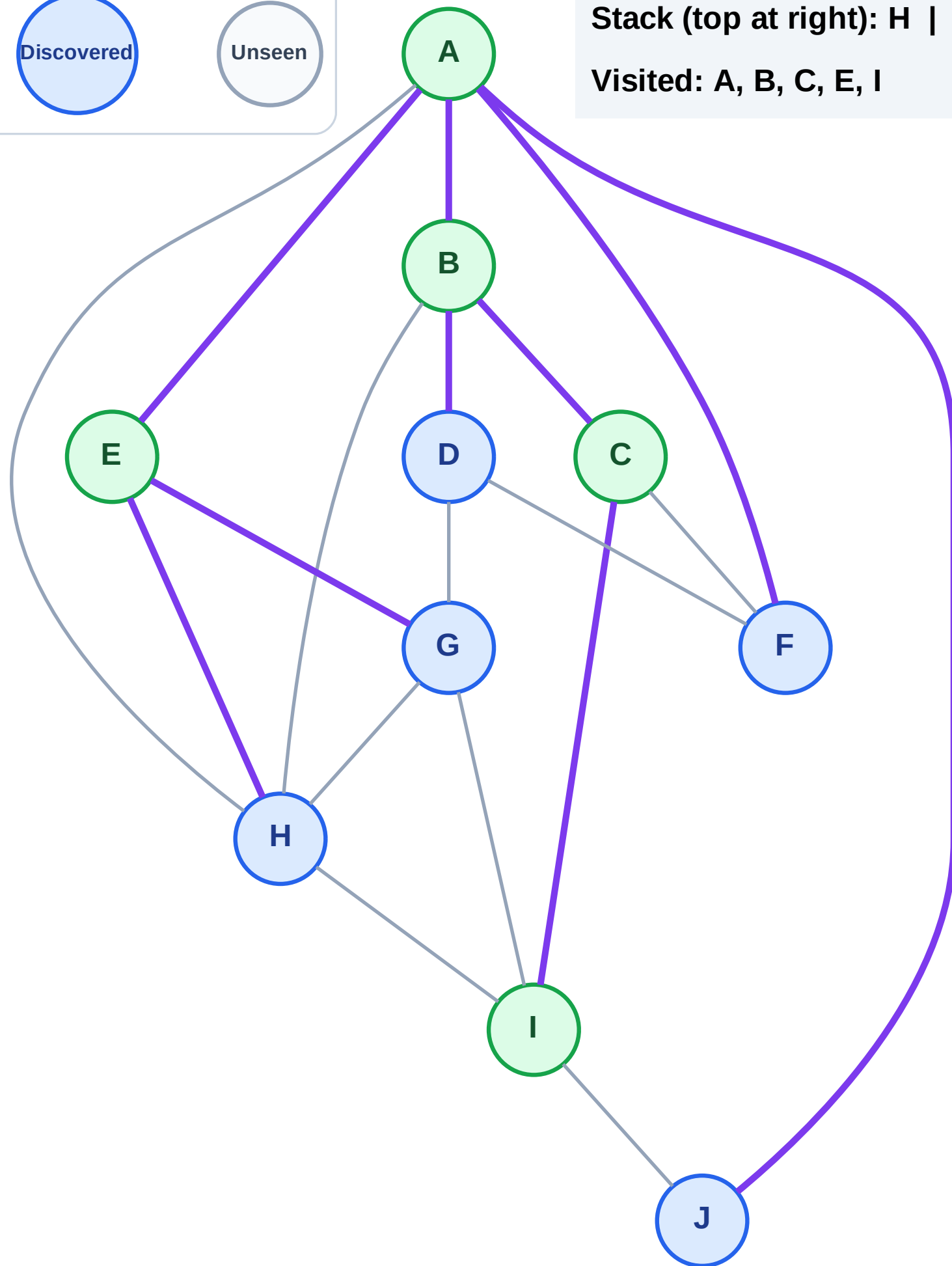
DFS Traversal

Step 11: No new neighbors from I



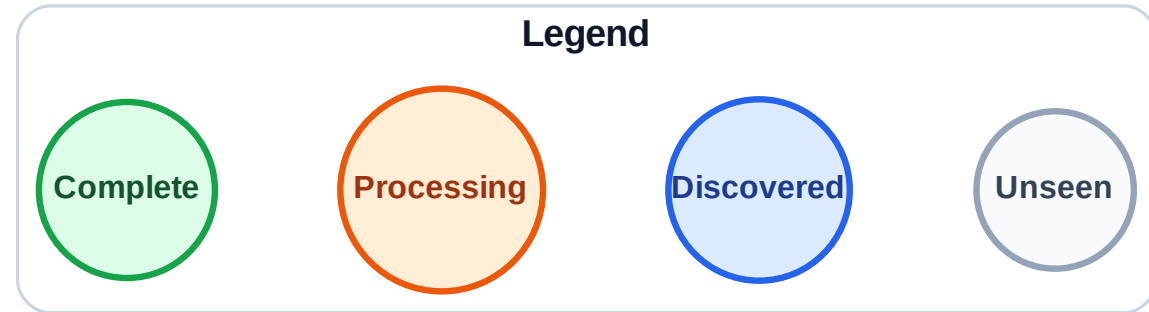
Stack (top at right): H | G | J | F | D

Visited: A, B, C, E, I



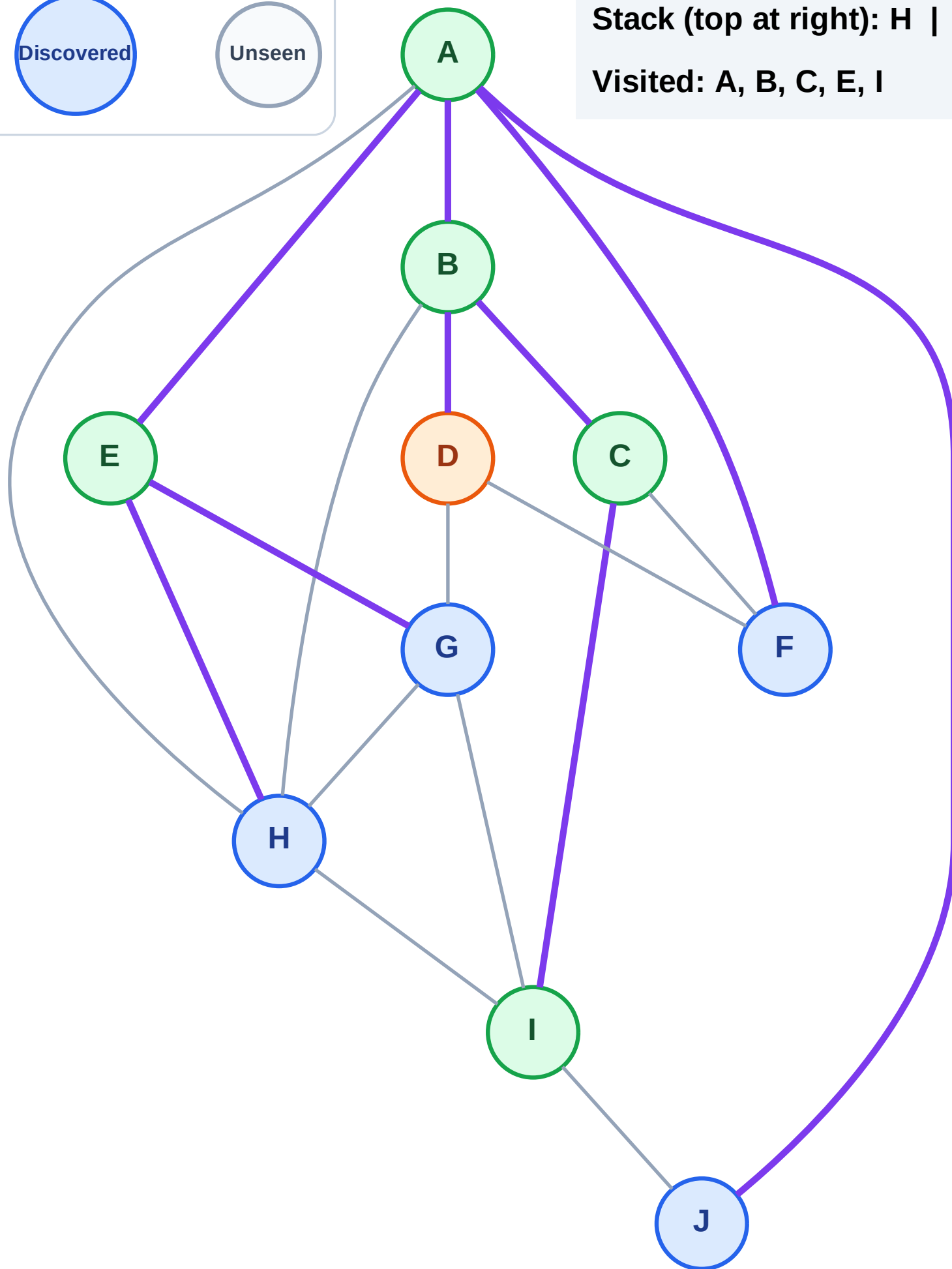
DFS Traversal

Step 12: Process node D



Stack (top at right): H | G | J | F

Visited: A, B, C, E, I



DFS Traversal

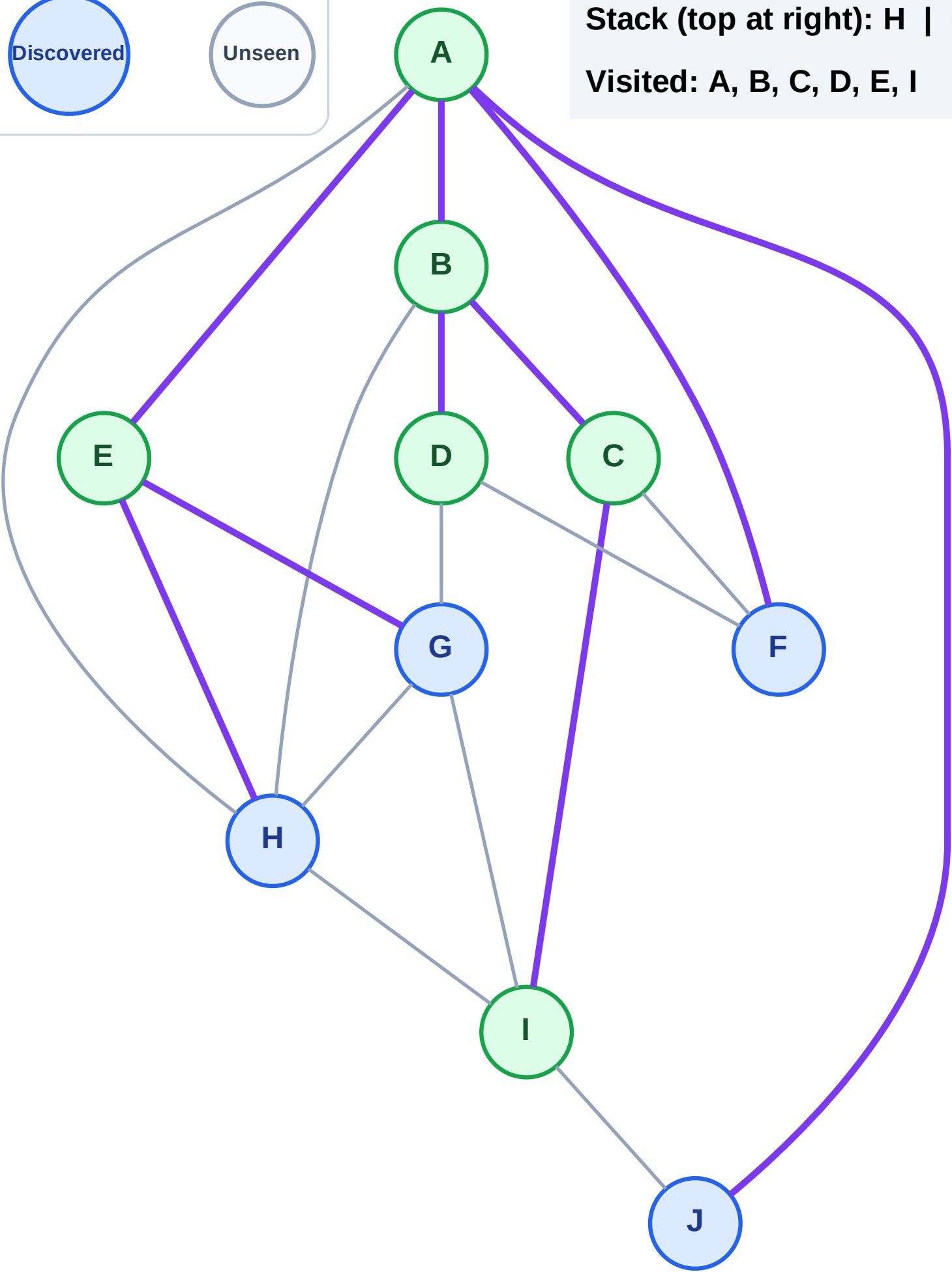
Step 13: No new neighbors from D

Legend



Stack (top at right): H | G | J | F

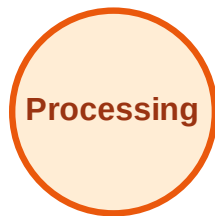
Visited: A, B, C, D, E, I



DFS Traversal

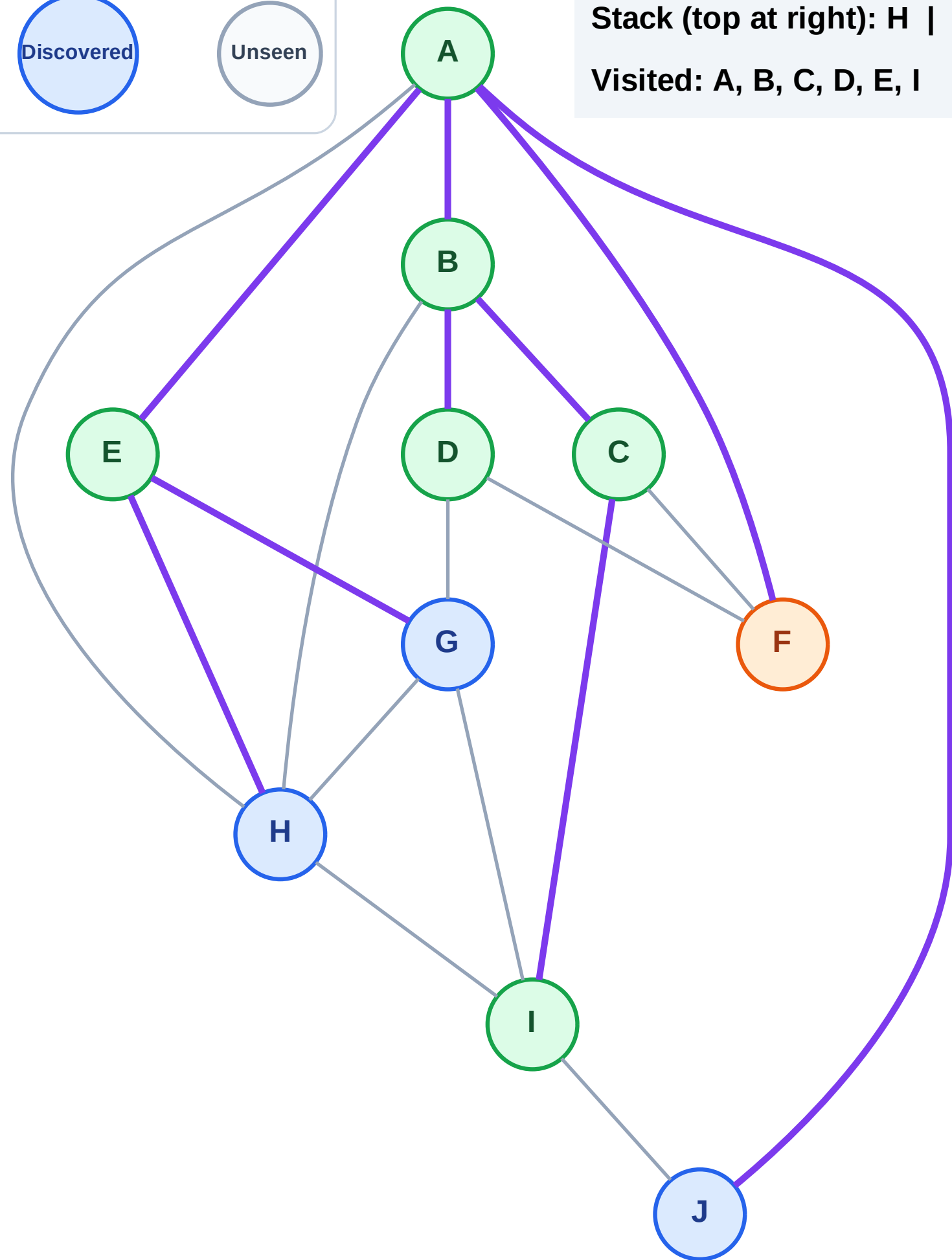
Step 14: Process node F

Legend



Stack (top at right): H | G | J

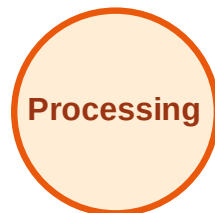
Visited: A, B, C, D, E, I



DFS Traversal

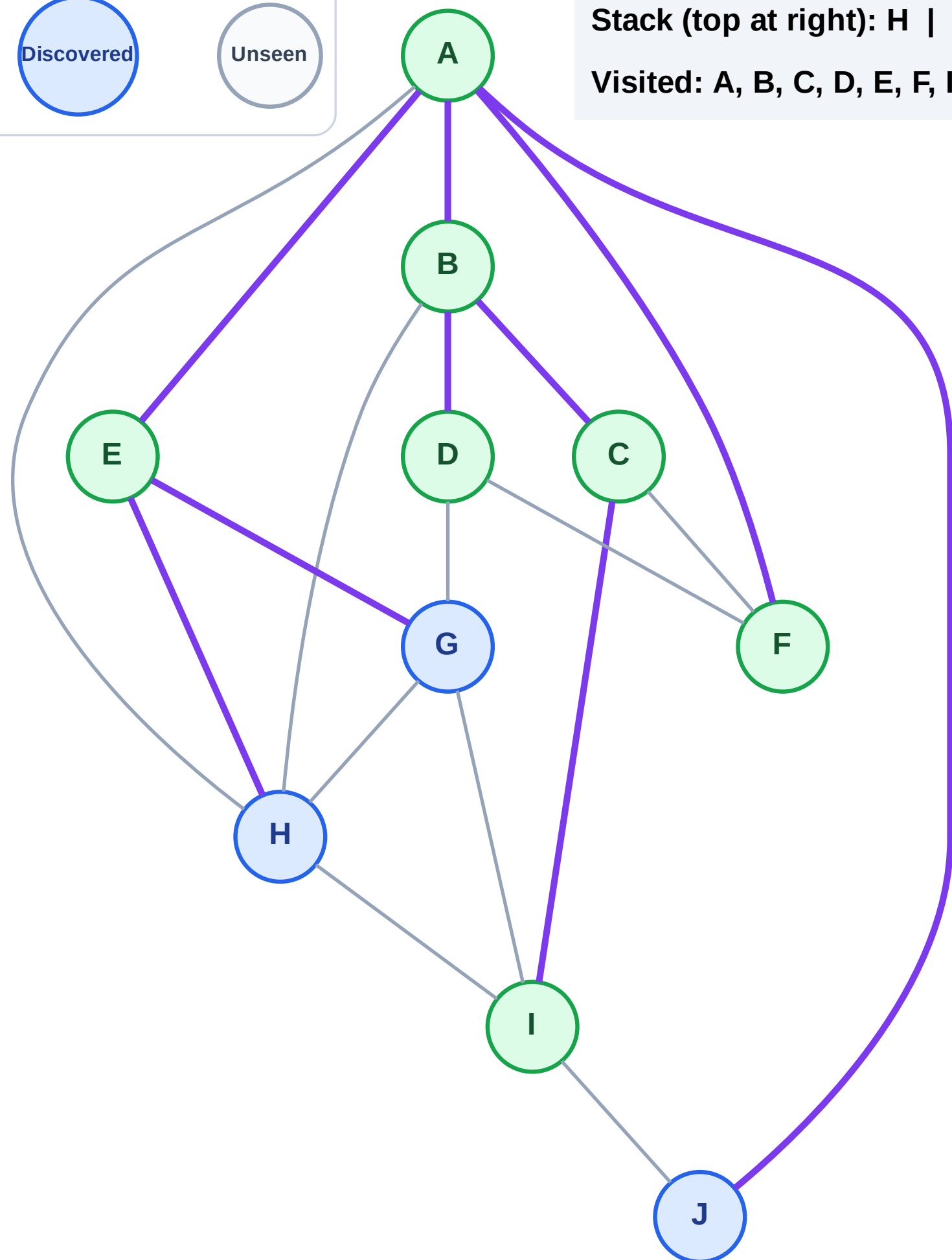
Step 15: No new neighbors from F

Legend



Stack (top at right): H | G | J

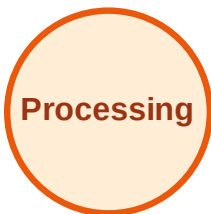
Visited: A, B, C, D, E, F, I



DFS Traversal

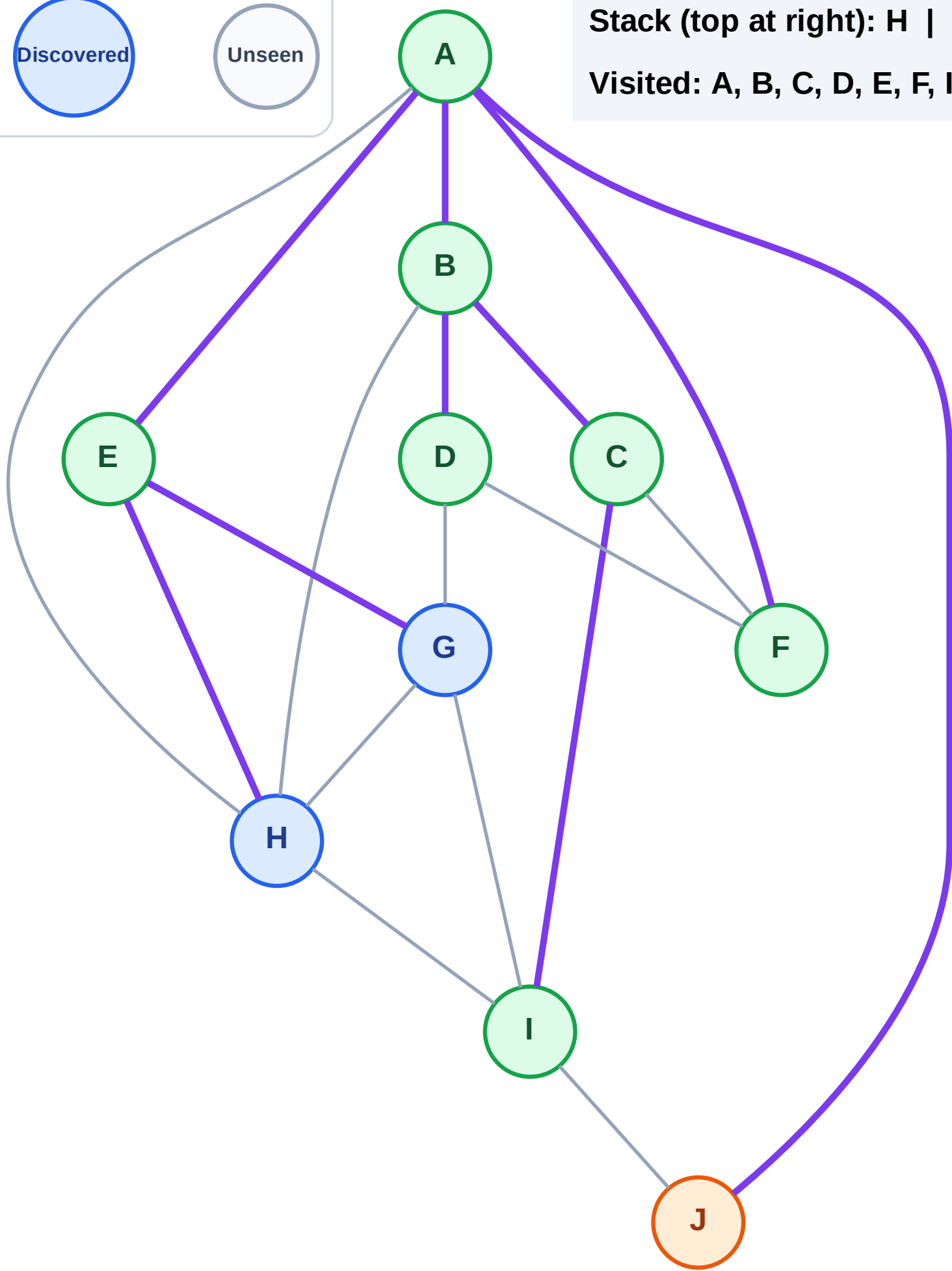
Step 16: Process node J

Legend



Stack (top at right): H | G

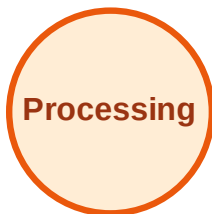
Visited: A, B, C, D, E, F, I



DFS Traversal

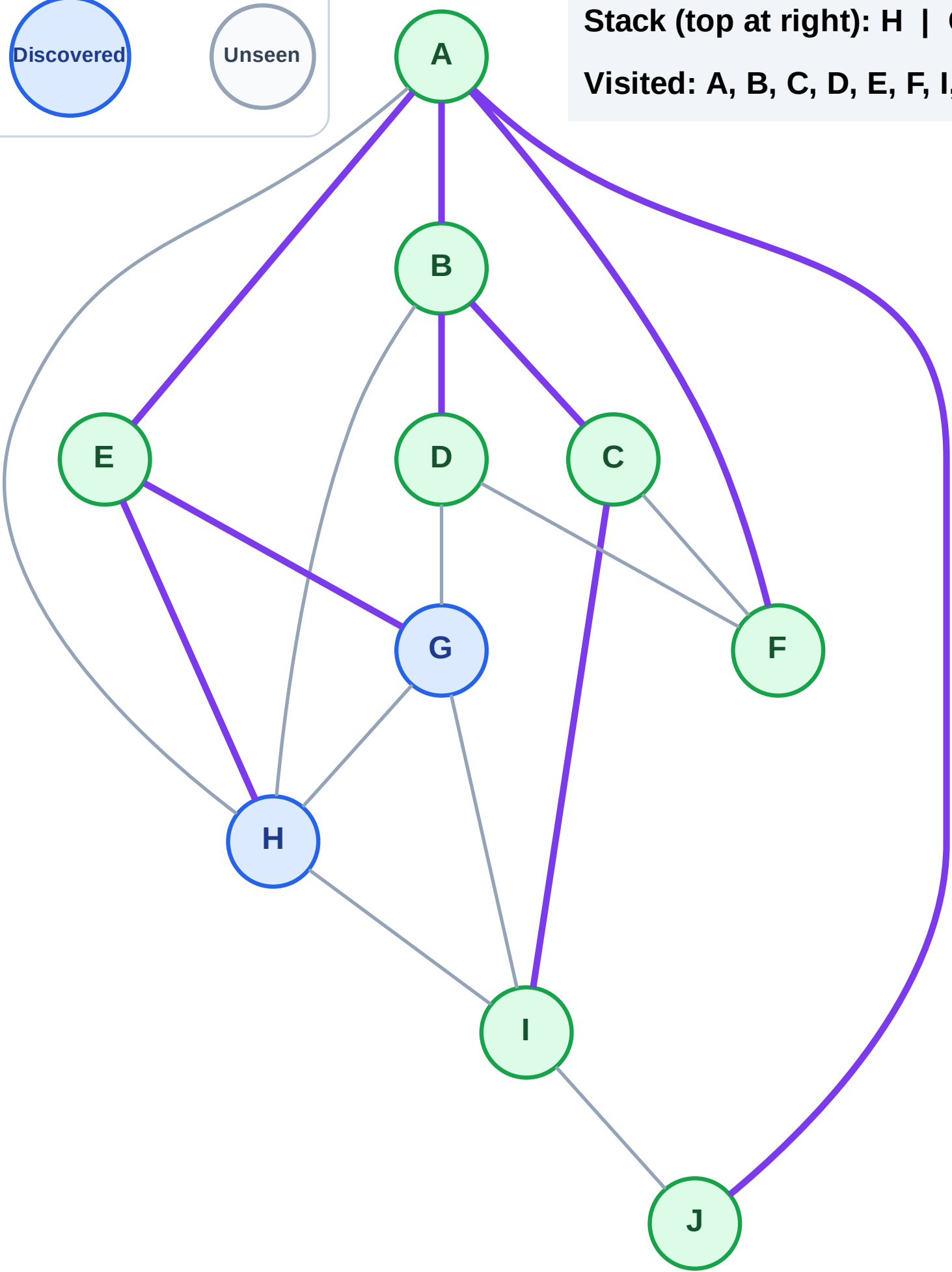
Step 17: No new neighbors from J

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

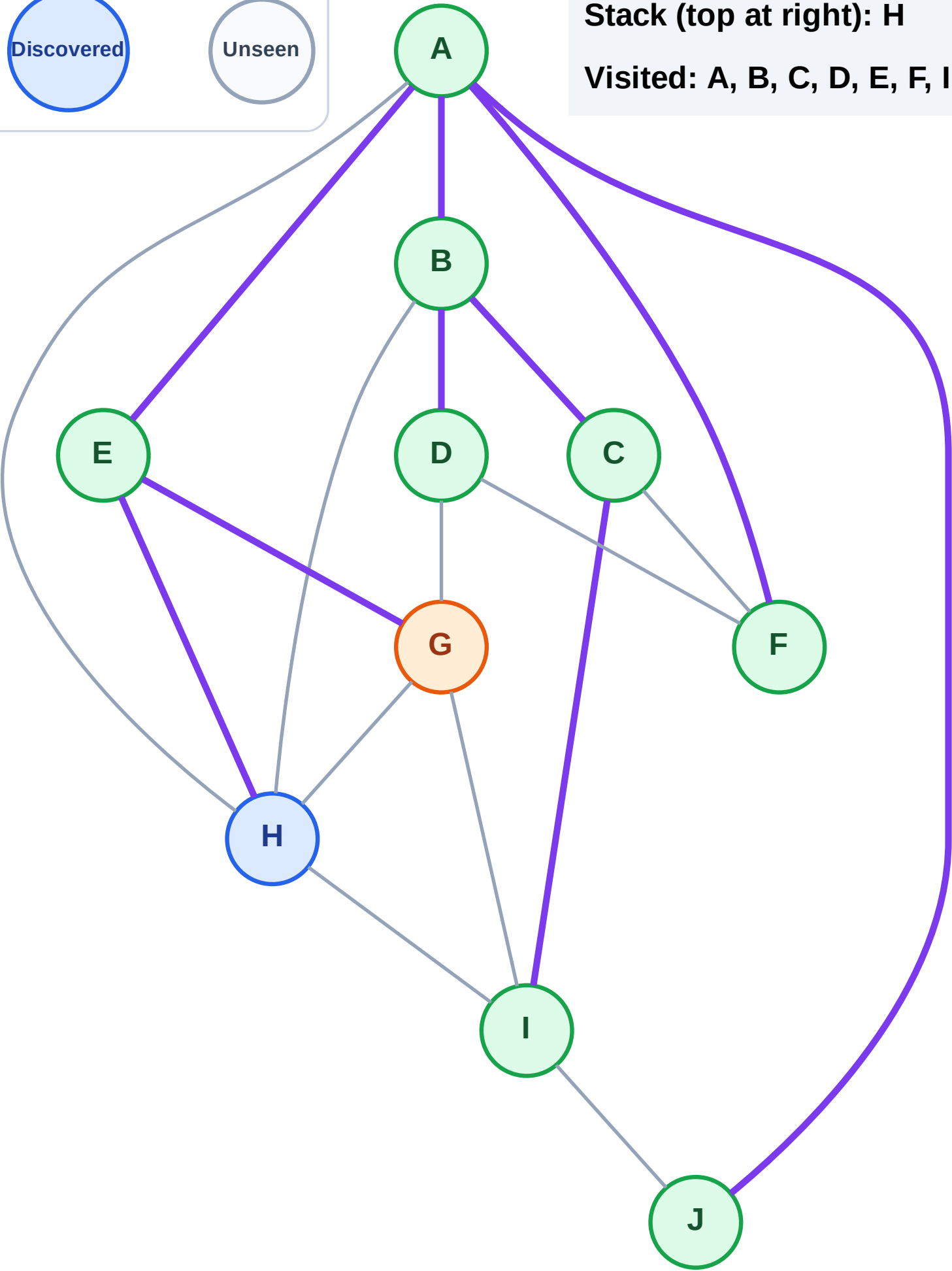
Complete

Processing

Discovered

Unseen

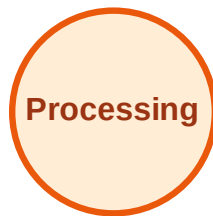
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

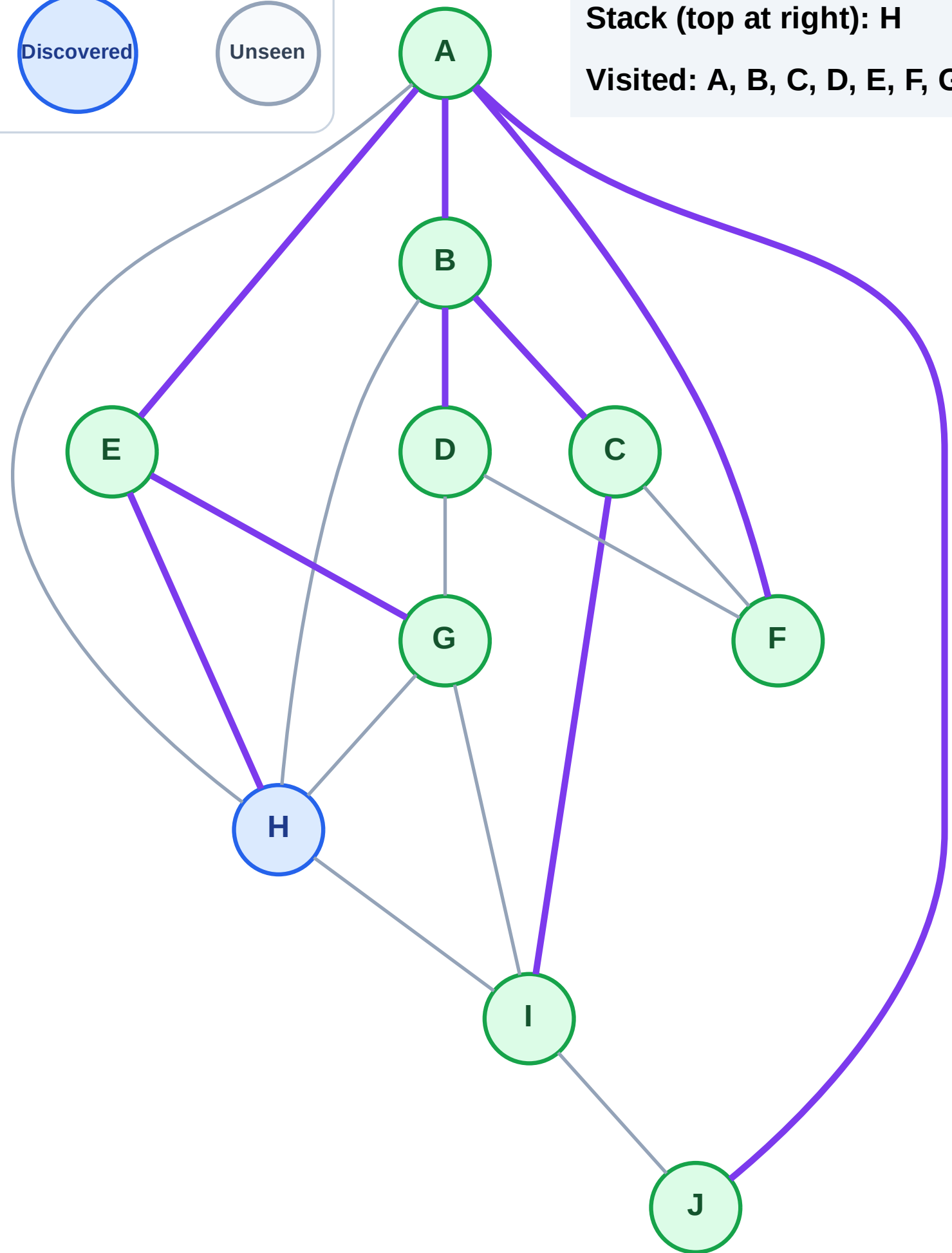
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

Complete

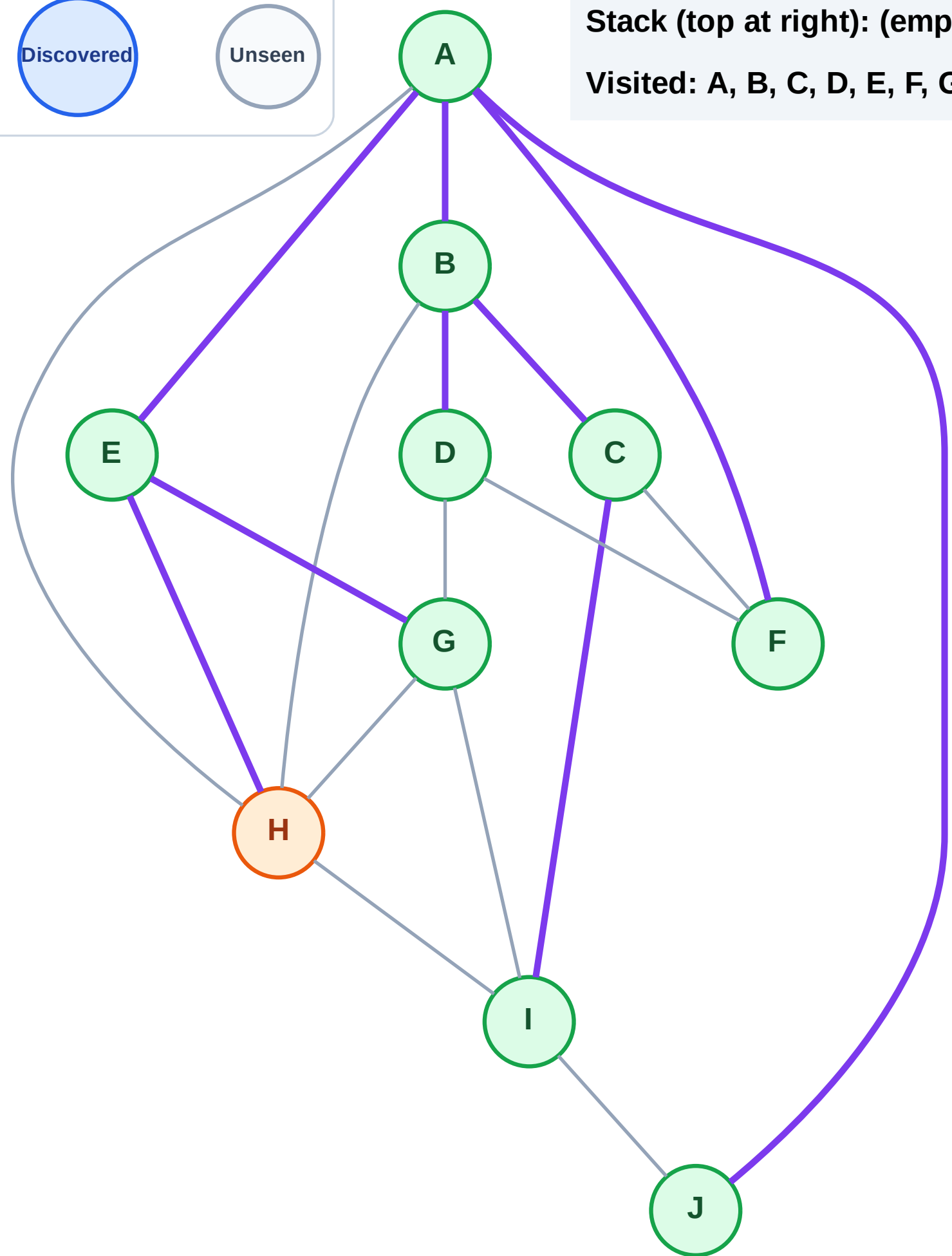
Processing

Discovered

Unseen

Stack (top at right): (empty)

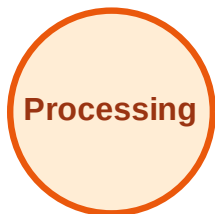
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

